



Machine Learning (After Midsem)

Generative Models

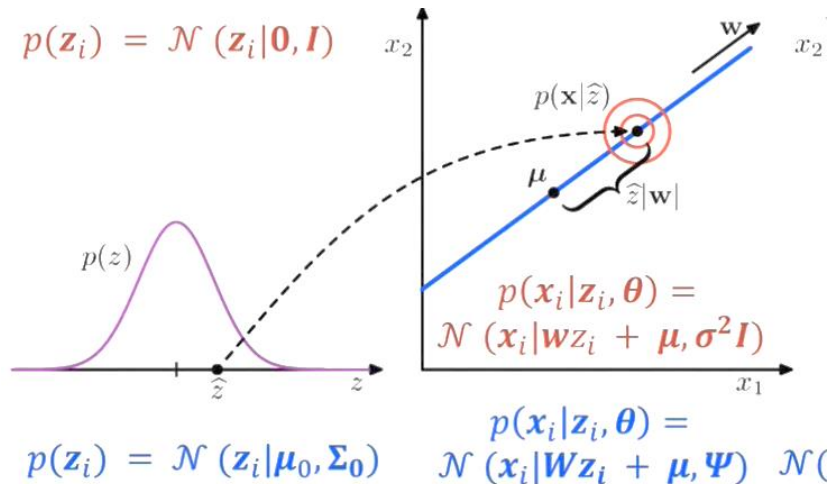


- Variational Autoencoder (VAE)
- Generative Adversarial Network (GAN)
- ...

Latent Variable Model



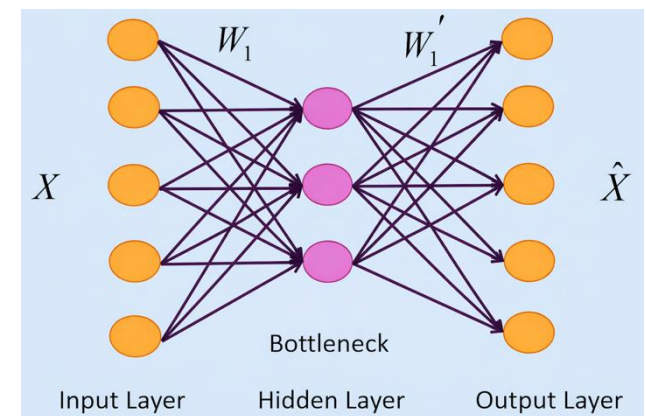
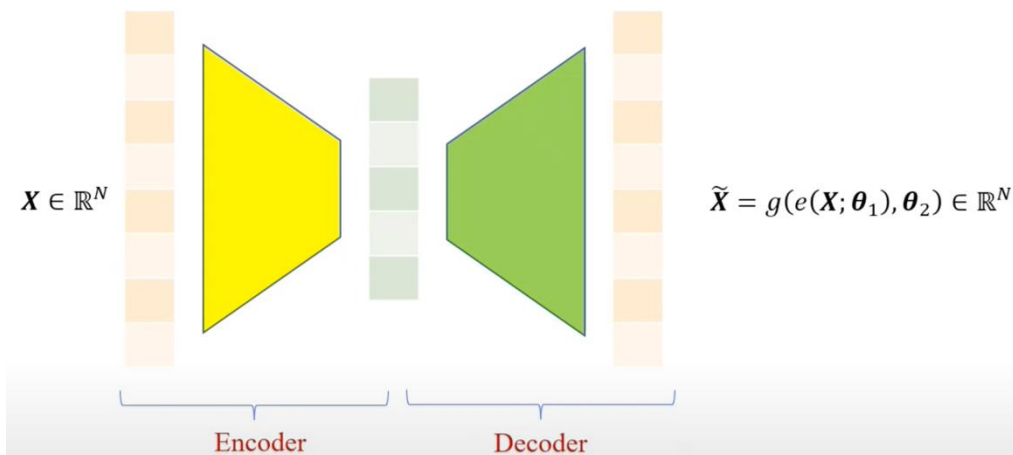
- Latent variable: $Z \rightarrow X$
- Prediction $p(X|Z)$
- Training $p(Z|X)$
- Learns low dimensional representation of a high dimensional data
- AE and VAE are examples of **latent variable model**



Autoencoder (VAE)



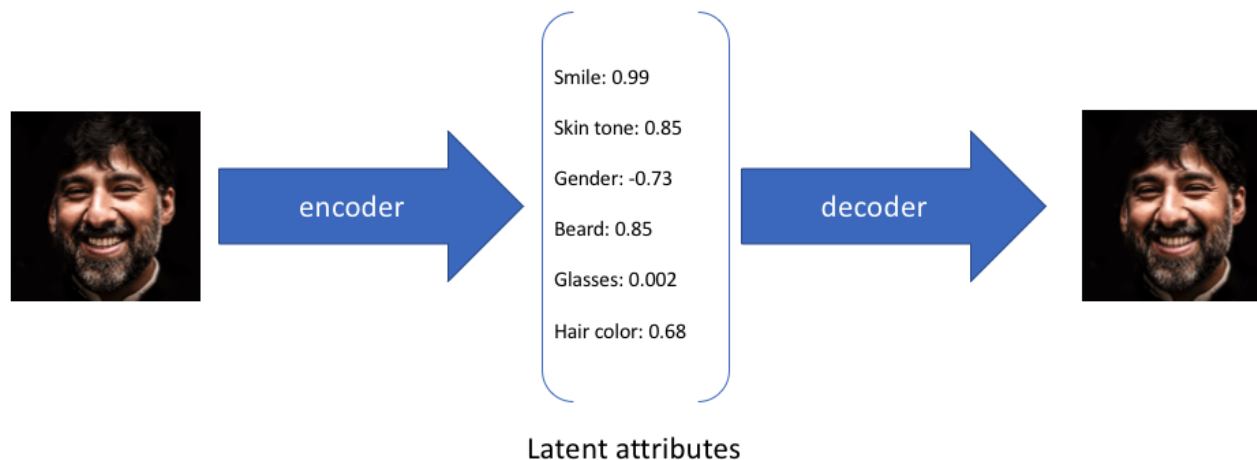
- Unsupervised Learning where Neural Networks are subject to the task of representation learning.
- Impose a bottleneck in the network.
- The bottleneck forces a compressed knowledge representation of the input.
- **Maps an input to a deterministic latent code and then tries to decode it to reconstruct the input.**
- Assumption is that a high degree of correlation/structure exists in the data.



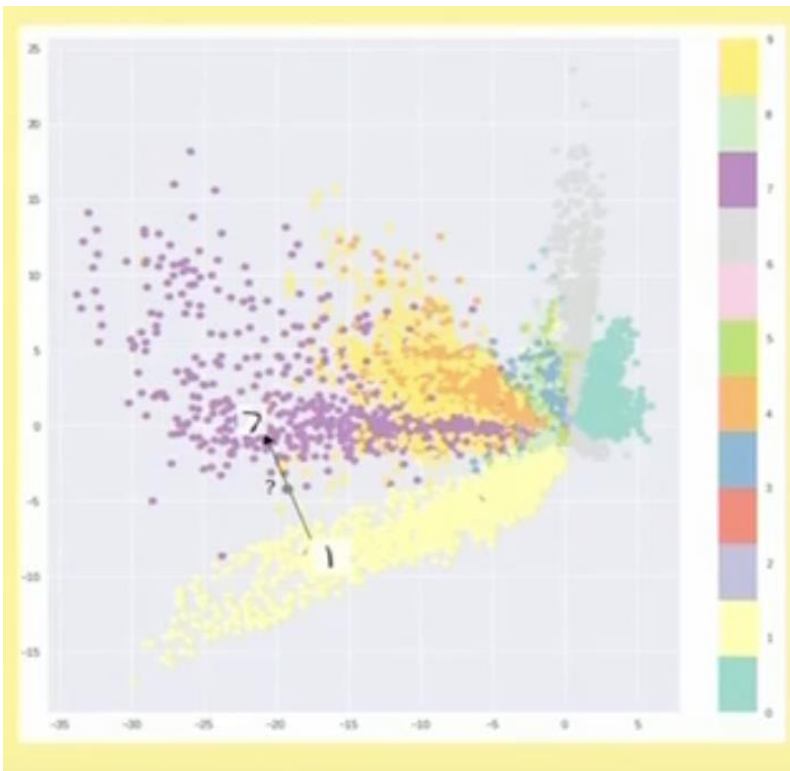
Autoencoder (AE)



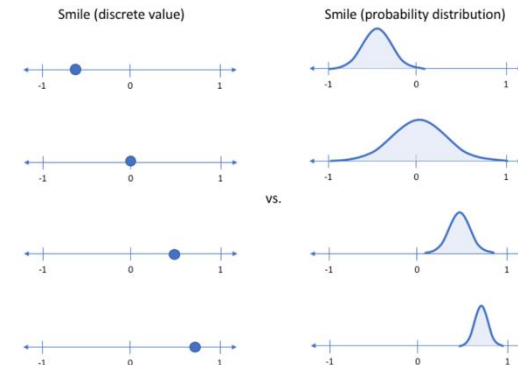
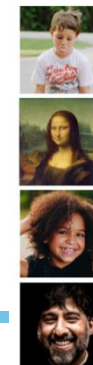
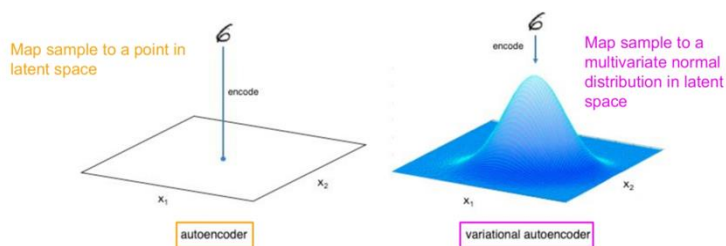
- Latent variable $Z \in R^M$ where $M \ll N$
- Idea is to learn encoder e and the decoder g such that
- $g^*, e^* = \arg \min_{g, e} \epsilon(X, \tilde{X})$
- Optimizing over function space is extremely difficult.
- Can find the reduced order representation but problem in generating new samples.
- Decoder needs a **distinct code** to generate a data point



Autoencoder (AE): Limitations



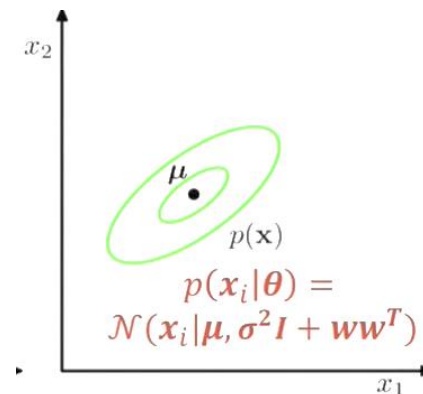
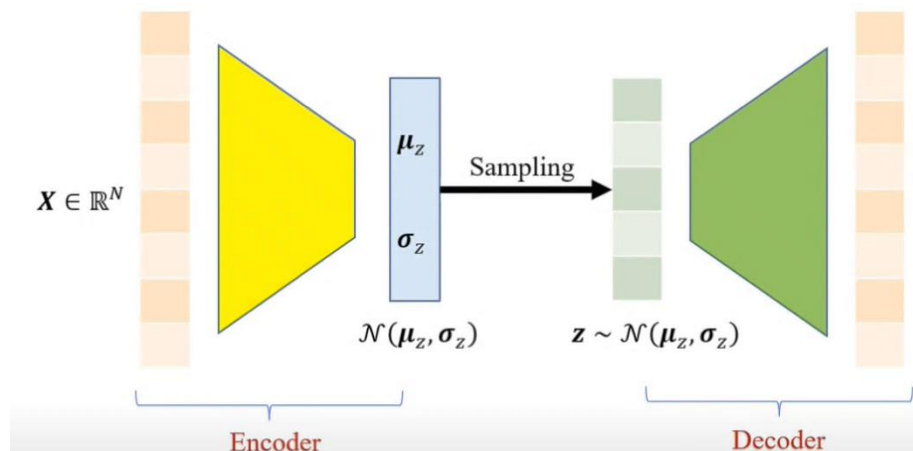
- Distinct clusters for each class.
- Decoder finds it difficult to generate new data since new code is needed for each new data.
- If we try to generate from such distinct clusters/discontinuous space, decoder will have to generate from unseen points, so generation will be unrealistic.
- **To overcome: Make latent space continuous yet class compact.**
- **Instead of a deterministic code, learn a distribution.**



VAE



- Latent space needs to **be regularized** such that it is continuous (yet maintain class specific compactness).
- Instead of encoding an input as a single point, we encode it as a distribution over the latent space.**
- VAE is therefore an autoencoder with having a regularized latent space.
- Change is only in the introduction of regularization
- Mean μ and standard deviation σ will come into picture
- μ controls where the encodings should be centered and σ controls the area to which the encodings vary from the center.



$$p(x_i | \theta) = \mathcal{N}(x_i | W\mu_0 + \mu, \Psi + W\Sigma_0 W^T)$$

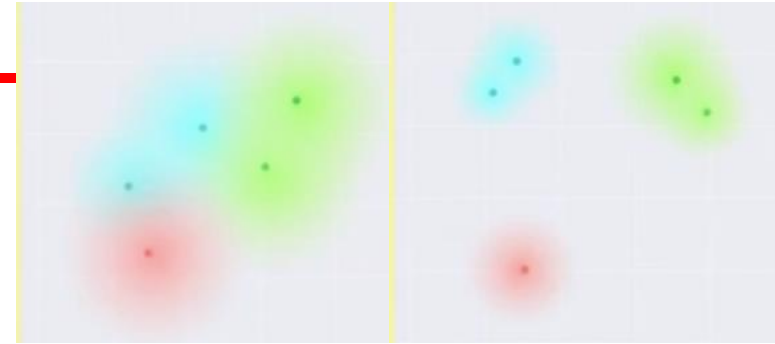
VAE



$$Loss \epsilon = \|X - \tilde{X}\|_2^2 + KL(\mathcal{N}(\mu_z, \sigma_z), \mathcal{N}(0, I))$$

class compactness
(similarity/dissimilarity of
data)

smooth
transition



we want smooth interpolation

Reconstruction Loss: Ensures that the decoded sample is close to the original input.

KL Divergence Loss: Regularizes the latent space to follow a standard normal distribution, encouraging a structured latent space. Ensures smooth transition.

Main steps:

1. Input encoded as distribution over the latent space
2. A point from the latent space is sampled from the distribution $z = \mu_z + \sigma_z \times \epsilon$ where $\epsilon \in \mathcal{N}(0, I)$
3. Sampled point is decoded and error is calculated
4. Error is backpropagated through the network

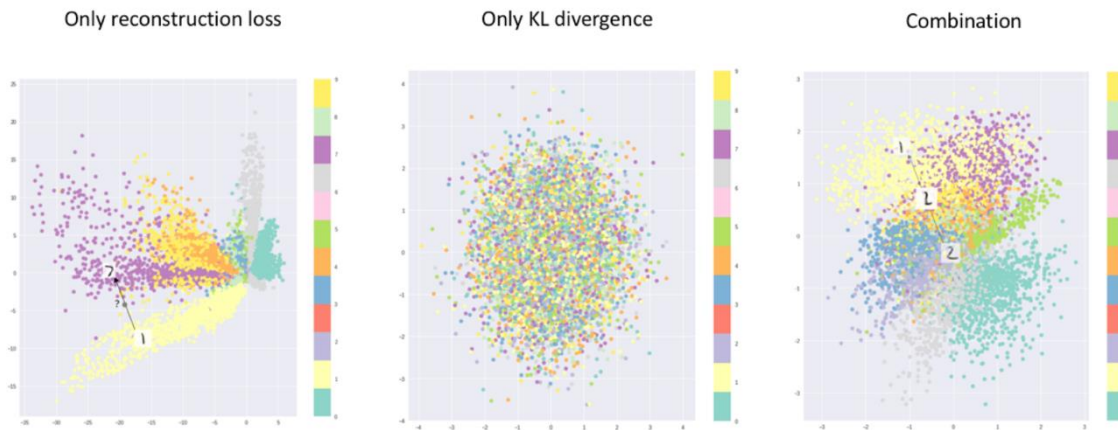
KL Divergence



- KL divergence between two probability distributions simply measures how much they diverge from each other. **If two distributions are very similar then KL is small.**
- $KL(q(Z|X) || p(Z|X)) = \sum_x q(x) \log_e \frac{q(x)}{p(x)} = E_q[\log_e q(x) - \log_e p(x)]$
- **Properties of KL Divergence**
- $KL(p||q) \text{ or } KL(q||p) \geq 0$
- $KL(p||q) \neq KL(q||p)$
- Compute $KL(P(x)||Q(x))$ of the following

x	0	1	2
P(x)	0.36	0.48	0.16
Q(x)	0.33	0.33	0.33

- Evaluate if KL is symmetric.

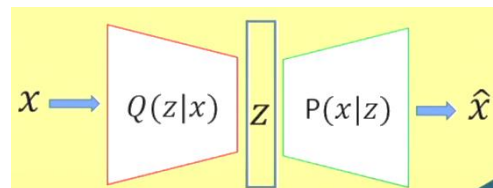


$$\|X - \tilde{X}\|_2^2 + KL(\mathcal{N}(\mu_z, \sigma_z), \mathcal{N}(0, I))$$

- Idea is $Z \rightarrow X$
- Baye's rule: $p(Z|X) = \frac{p(X|Z)p(Z)}{p(X)}$, here $p(X) = \int p(X|Z)p(Z) dZ$
- **Integral term makes it difficult as Z is multi-dimensional (intractable).**
- Variational Inference
- To manage, we assume $p(Z|X) = q(Z|X)$, where **q is a tractable distribution** (Gaussian, for example, so that it has a mean and variance).
- So, **minimize the gap** between $p(Z|X)$ and $q(Z|X)$ using **KL divergence**.

Variational Lower Bound

- **KL divergence**, $KL(q(Z|X) || p(Z|X)) = \sum_z q(Z|X) \log_e \frac{q(Z|X)}{p(Z|X)}$
- Can be simplified to $KL = -\sum_z q(Z|X) \log \frac{p(X|Z)p(Z)}{q(Z|X)} + \log p(x)$
- or $\log p(x) = \sum_z q(Z|X) \log \frac{p(X|Z)p(Z)}{q(Z|X)} + KL$. The red part is the Variational Lower Bound
- As $KL \geq 0$, so $\sum_z q(Z|X) \log \frac{p(X|Z)p(Z)}{q(Z|X)} \leq \log p(x)$.
- A lower bound is found, so we need to maximize $p(x)$
- Same as $\max \sum_z q(Z|X) \log \frac{p(X|Z)p(Z)}{q(Z|X)}$
- $\max \sum_z q(Z|X) \log \frac{p(X|Z)p(Z)}{q(Z|X)} = \max \sum_z q(Z|X) \log p(X|Z) + \sum_z q(Z|X) \log \frac{p(Z)}{q(Z|X)}$



class compactness

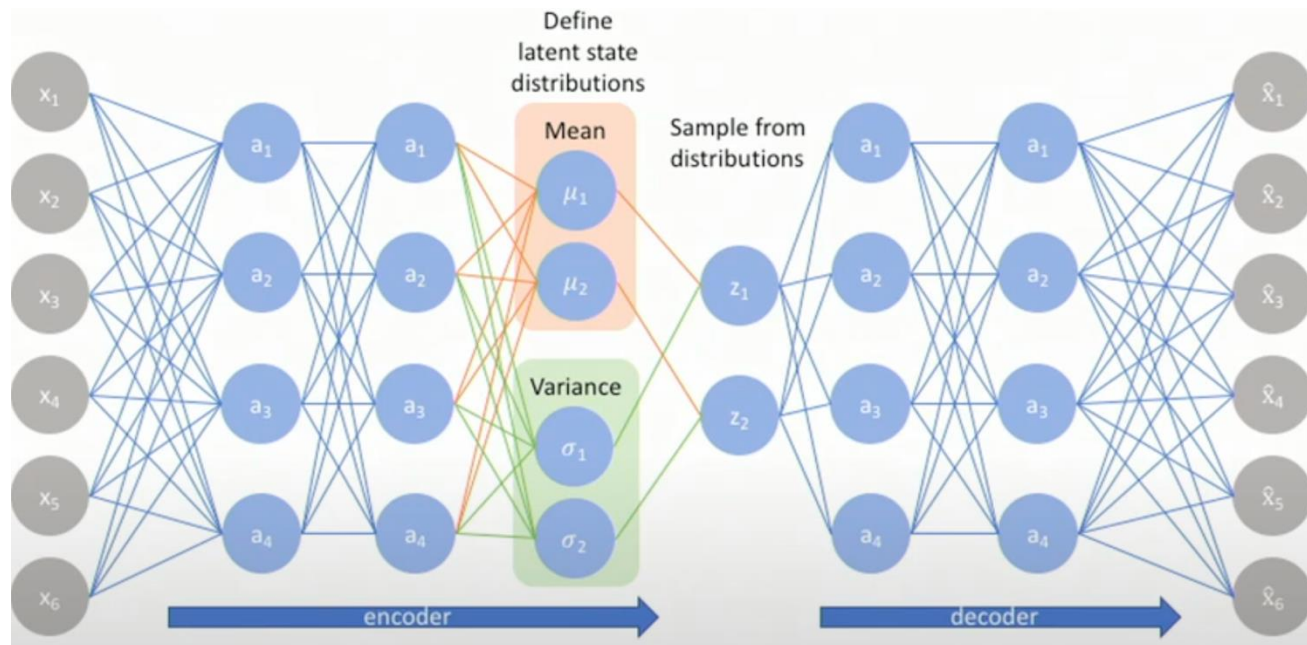
$-KL(q(Z|X)||p(Z))$

KL Divergence



- Loss of VAE $\sum_z q(Z|X) \log p(X|Z) - KL(q(Z|X)||p(Z))$
- KL part can be simplified to $-\frac{1}{2} \sum_i (1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2)$

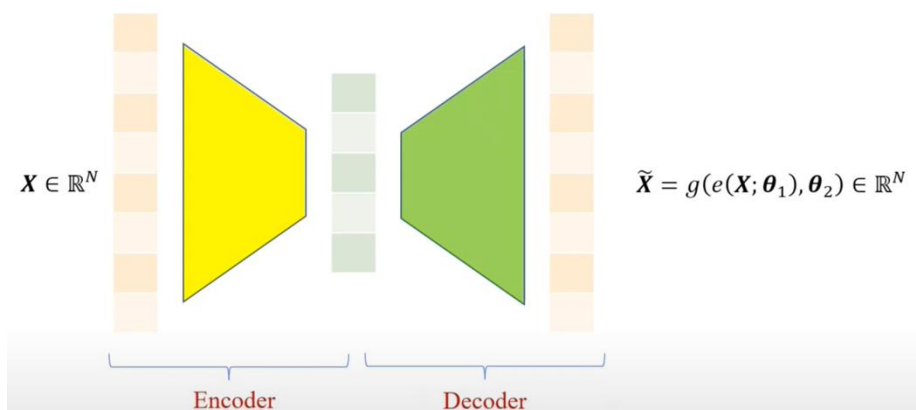
$KL(q(Z|X)||p(Z))$. Here $p(Z) = N(0, I) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$ and $q(Z|X) = N(\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$



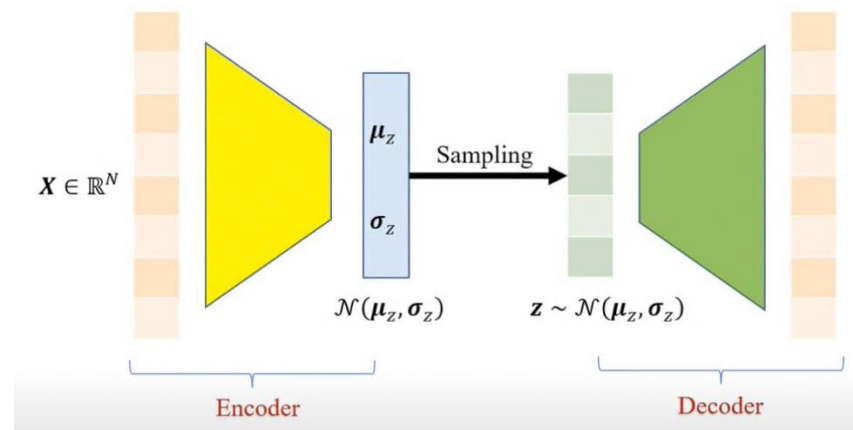
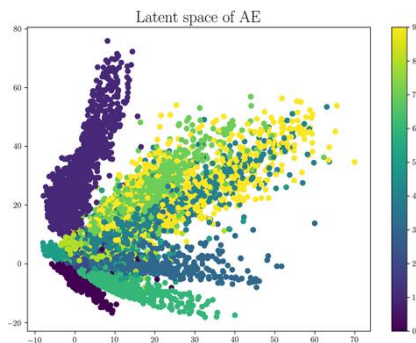
-
- Diagram illustrating the reparameterization trick:
- Legend:**
 - Gray Diamond: Deterministic node
 - Blue Circle: Random node
 - Left Side (Original Model):**
 - Encoder model (deterministic) takes input x and outputs parameters μ and σ .
 - Latent variable z is sampled from the distribution $q(z|x)$.
 - Decoder model (deterministic) takes z as input.
 - Right Side (Reparameterized Model):**
 - Encoder model (deterministic) takes input x and outputs parameters μ and σ .
 - Latent variable z is reparameterized as $z = \mu + \sigma \odot \epsilon$, where $\epsilon \sim N(0, 1)$ is a standard normal random variable.
 - Decoder model (deterministic) takes z as input.
 - Transformation:** A blue arrow labeled "reparameterization" points from the left model to the right model.



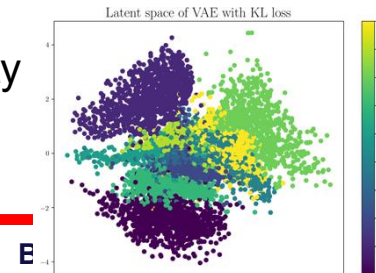
AE vs VAE



- Deterministic
- $Loss \epsilon = \|X - \tilde{X}\|_2^2$
- Non-regularized latent space
- Poor generative capability



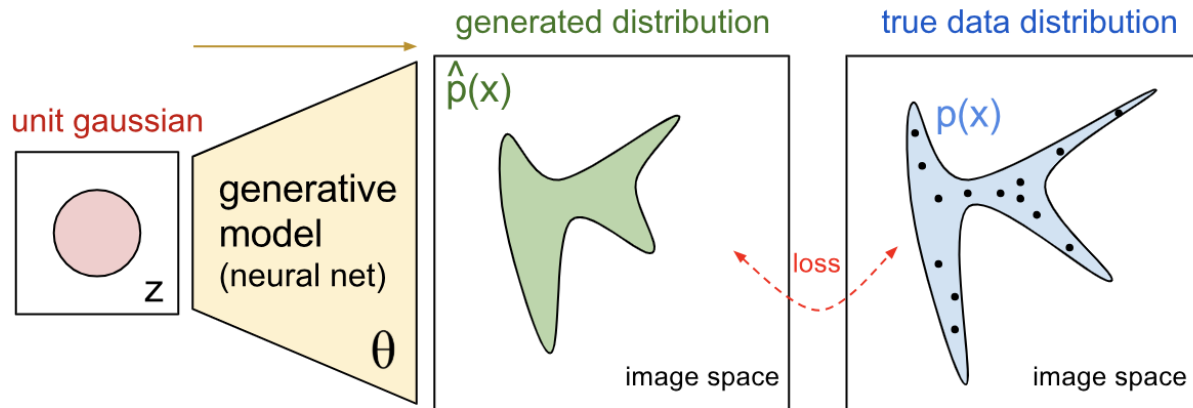
- Probabilistic $p(X|Z) = \frac{p(X|Z)p(Z)}{p(X)}$
- $Loss \epsilon = \|X - \tilde{X}\|_2^2 + KL(\mathcal{N}(\mu_z, \sigma_z), \mathcal{N}(0, I))$
- KL part: $-\frac{1}{2} \sum_i (1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2)$
- Regularized latent space as it follows a prior (class compactness and smooth transition)
- Good generative capability



Generative Adversarial Network (GAN)



- Implicitly defines a probability distribution
- Sample a code vector z from a simple distribution (Gaussian, Uniform etc.)
- Feed it to a generator network, which is trained to map z to x .

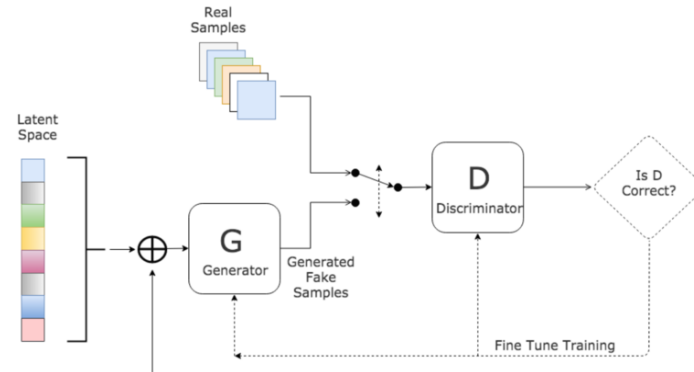


- How to make these two distributions similar?
- $KL(p(x)||\hat{p}(x))$

GAN Architecture

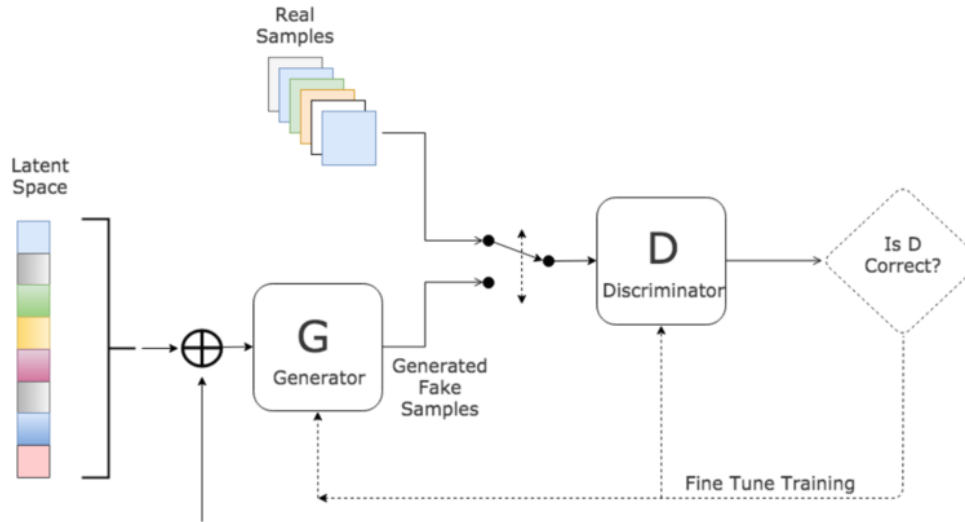


- Two major networks, Generator (G) and Discriminator (D)
- Competes to play a zero-sum game to generate realistic images.
- G samples a $\mathbf{z}$ vector and tries to produce realistic image.
- D tries to distinguish fake sample (from G) and real samples.



- If $D(x)$ is probability of belonging to real class of a given sample x .
- D will try to increase $D(x)$ for real samples and decrease $D(x)$ for fake/generated samples. $D \in [0,1]$
- G will try to increase $D(x)$ for its own generated samples

GAN Loss Function



Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

for k steps **do**

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m [\log D(x^{(i)}) + \log(1 - D(G(z^{(i)})))]$$

end for

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log(1 - D(G(z^{(i)})))$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))].$$

Discriminator
Loss

$$\overline{E_x [\log(D(x))] + E_z [\log(1 - D(G(z)))]}$$

Generator
Loss

$$\mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))].$$

Generative Adversarial Network (GAN)



- Discriminator and Generator are going in opposite directions
- In other words, the generator is trying to fool the discriminator to generate realistic samples.
- **The training happens one after the other**
- First Generator generates some samples from the latent input z
- Based on this, Discriminator learns, then Generator trains itself
- Steps:
 - Define GAN architecture based on the application
 - Train discriminator to distinguish real vs fake data
 - Train the generator to fake data that can fool the discriminator
 - Continue discriminator and generator training for multiple epochs
 - Save generator model to create new, realistic fake data

VAE & GAN



VAE

$$\max \sum_z q(\mathbf{Z}|\mathbf{X}) \log p(\mathbf{X}|\mathbf{Z}) - KL(q(\mathbf{Z}|\mathbf{X})||p(\mathbf{Z}))$$

- Started from $\min KL(q(\mathbf{Z}|\mathbf{X})||p(\mathbf{Z}|\mathbf{X}))$ and Bayes Theorem.

- We got $\log p(\mathbf{x}) = \sum_z q(\mathbf{Z}|\mathbf{X}) \log \frac{p(\mathbf{X}|\mathbf{Z})p(\mathbf{Z})}{q(\mathbf{Z}|\mathbf{X})} + KL$

- The red part is the Variational Lower Bound

- Now we end at $\max \sum_z q(\mathbf{Z}|\mathbf{X}) \log \frac{p(\mathbf{X}|\mathbf{Z})p(\mathbf{Z})}{q(\mathbf{Z}|\mathbf{X})}$

- Expanding this will give the loss function

GAN

$$\min_G \max_D E_{x \sim p_{data}(x)} [\log(D(x))] + E_{z \sim p(z)} [\log(1 - D(G(z)))]$$

DT using VAE and GAN



Contents lists available at [ScienceDirect](#)

Mechanical Systems and Signal Processing

journal homepage: www.elsevier.com/locate/ymssp



Download the paper
and read

Deep digital twins for detection, diagnostics and prognostics

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^a Centre for Asset Integrity Management, University of Pretoria, South Africa

^b Kriterion AI Research, South Africa



- Solving PHM as a supervised classification problem may not be the best approach
- A deep digital twin (DDT) is proposed
- Trying to **learn the distribution** of the healthy data
- For new data, you can now check if it is **from that distribution**
- If not from the same distribution, then it must be faulty data
- **Doesn't rely on historical faulty data**
- Uses VAE and GAN to build the twin

DT using VAE and GAN



- Uses **conditional VAE, GAN**
- $p(X|Y, Z)$, where Y is the condition. Previously we were solving for $p(X|Z)$
- In this work, if we learn the distribution $p(X|Y, Z)$ then for a new condition Y^* we must be able to predict $p(X|Y^*, Z)$
- $p(X|Y^*, Z)$ will basically tell you how the asset is behaving.
- The goal of the DDT training is to learn some approximation $q(X|Y, Z)$ which closely matches the true distribution $p_{data}(X|Y, Z^*)$
- To develop health indicator that reflects the behaviour of the asset.
- Uses likelihood (VAE) and likelihood ratio (GAN) as the health indicators.

DT using VAE and GAN



$$p_{\theta}(\mathbf{z}|\mathbf{x}, \theta^{\star}) = \frac{p_{\theta^{\star}}(\mathbf{x}|\mathbf{z})p_{\theta}(\mathbf{z})}{p_{\theta^{\star}}(\mathbf{x})} \quad p_{\theta}(\mathbf{x}) = \int \mathbf{p}_{\theta}(\mathbf{x}|\mathbf{z})\mathbf{p}_{\theta}(\mathbf{z}) \, d\mathbf{z}.$$

$$\mathcal{L}_{VAE} = ||g_{\theta}(f_{\phi}(\mathbf{x})) - \mathbf{x}||_2 + \sum_i^N \frac{1}{N} \sum_d^{D_{latent}} \mu_{x_{id}}^2 + \sigma_{x_{id}}^2 - \log(\sigma_{x_{id}}) - 1,$$

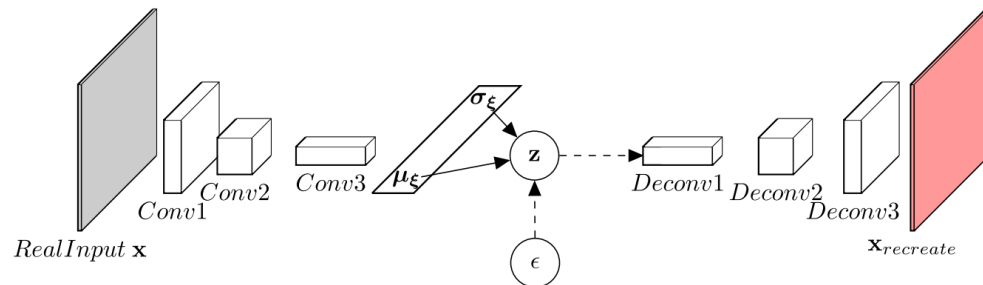


Fig. 1. Basic architecture of a Variational Autoencoder (VAE) with convolutional and de-convolutional layers, dashed lines denote sampling from a stochastic node. On the left side of the graph is the encoder network $q_{\xi}(\mathbf{z}|\mathbf{x}, \phi)$; On the right is the decoder network $p_{\theta}(\mathbf{x}|\mathbf{z})$.

$$HI_{VAE} = p_{\theta}(\mathbf{x}).$$

DT using VAE and GAN

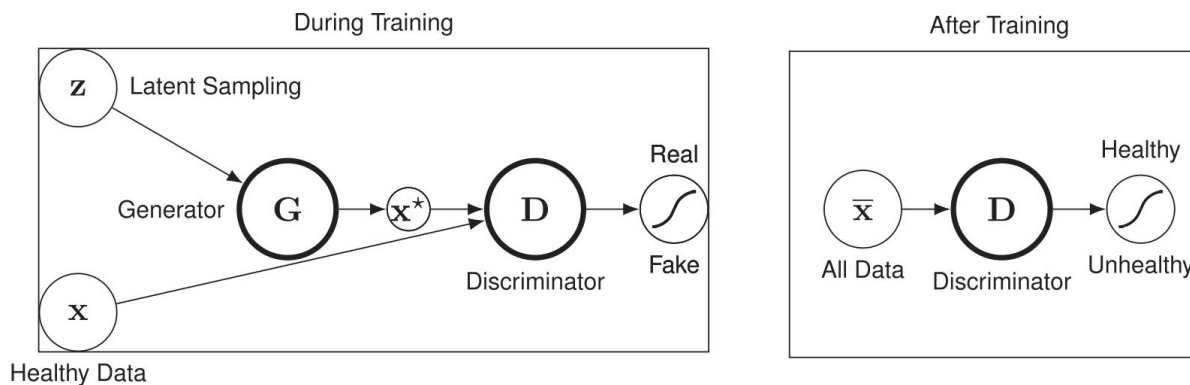
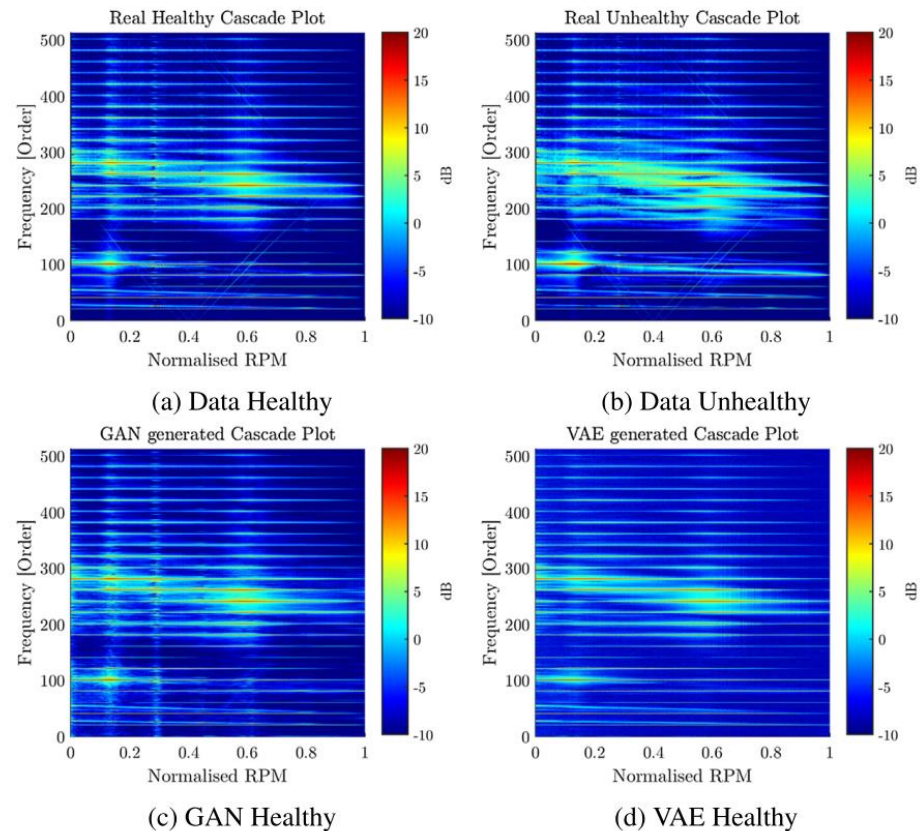
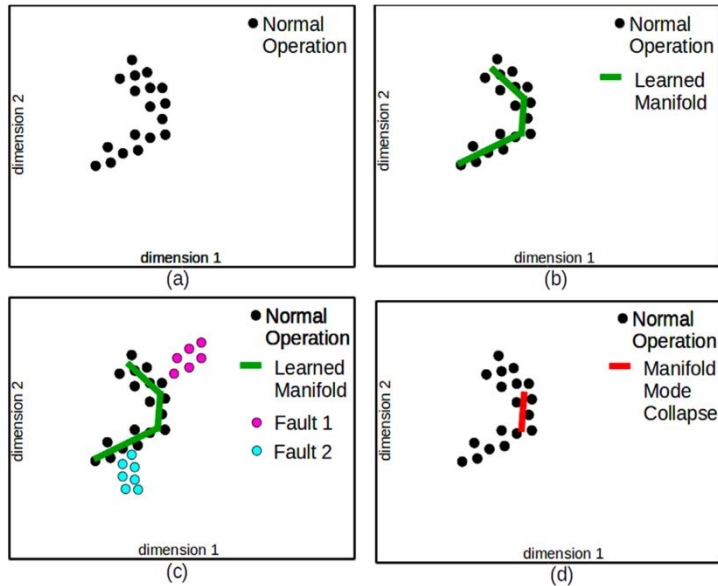


Fig. 2. Diagram demonstrating the GAN DDT. The left depicts the GAN training process and the right image demonstrates the process of producing a health indicator is shown.

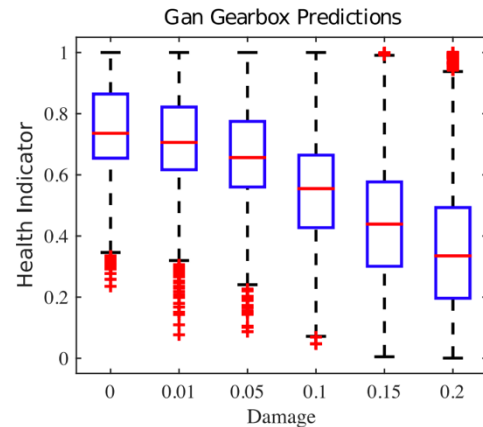
$$HI = D(\mathbf{x}).$$

DT using VAE and GAN

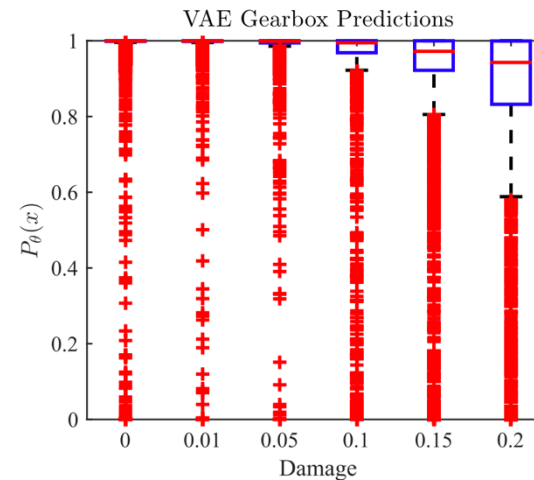


- Learned distributions used for anomaly detection
- Mode collapse

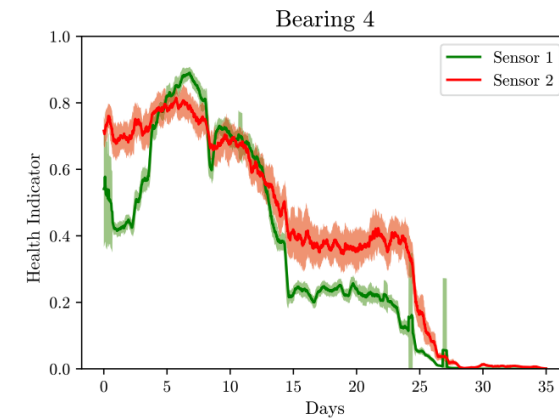
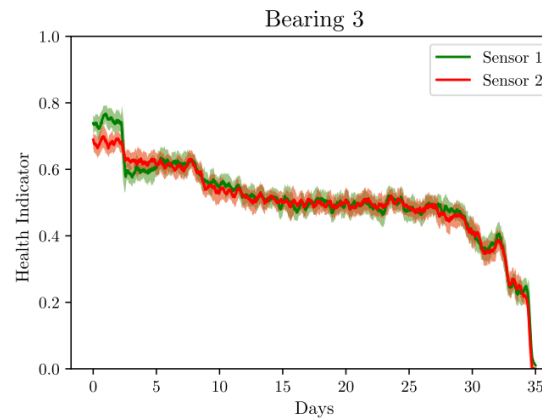
Gearbox and Bearing Monitoring



(a) GAN



(b) VAE



- HI value is continuously decreasing with increase in damage severity.
- GAN outperforms VAE.



BITS Pilani
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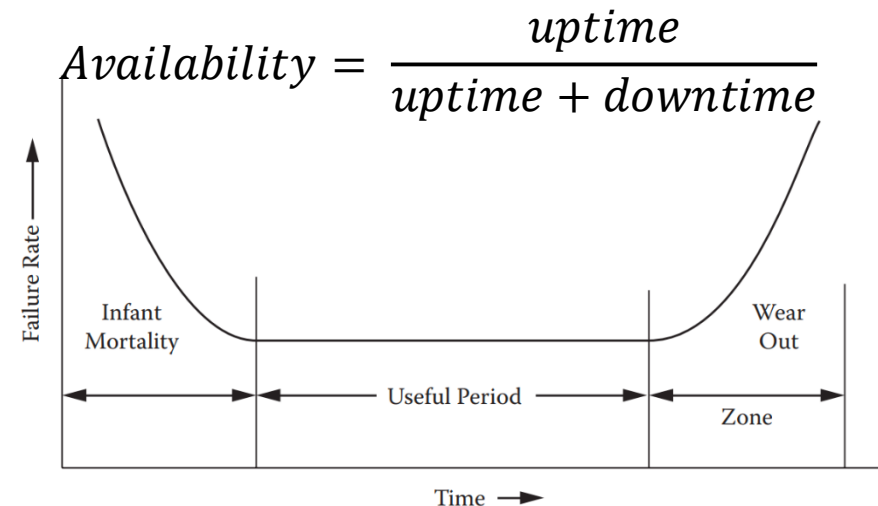
Predictive Maintenance

Introduction



- Machines have to be maintained to avoid failures.
- All over the world, three different types of maintenance techniques are adopted, **reactive maintenance**, **preventive maintenance**, and **predictive maintenance**.
- The benefits of planned maintenance are as follows:

- Eliminate unnecessary maintenance
- Reduce rework costs
- Reduce lost production caused by failures
- Reduce repair parts inventory
- Increase process efficiency
- Improve product quality
- Extend the operating life of plant systems
- Increase production capacity
- Reduce overall maintenance costs
- Increase overall profit



Bath tub curve: The typical machine failure rate versus time plot in the shape of a bath tub.

Aim is always to increase the machine's availability

Types of Maintenance



- **Reactive maintenance** means that we react to a need for maintenance of the machine, because the machine has failed completely. Also known as breakdown maintenance.
- In **preventive maintenance**, the maintenance on a particular machine is done in a regular periodic manner at a fixed frequency. Also known in periodic maintenance.
- **Predictive maintenance** is done on a machine depending on its need. The decision to conduct maintenance or not to conduct maintenance depends upon the machine's past and present condition.
- **Machinery condition monitoring** deals with the maintenance aspects of machines based on the present and past condition of the machine.

Failure Modes Effects and Criticality Analysis (FMECA)



- FMECA is a methodology widely used in the industry to identify and analyze all potential failure modes of the various parts of a system, determine the effects these failures may have on the system, and how to avoid the failures or mitigate the effects of the failures on the system.
- FMECA in Machine Maintenance: It allows analysis early in the planning stage of maintenance; lets one determine the critical machines that will require more attention than the rest.
- Risk Priority Number (RPN) = $O * S * D$
- O: The rank of occurrence of a failure mode; S: The rank of severity of the failure mode; D: The rank that a failure will be detected before the system reaches end-user.
- O, D, and S are generally set on a scale of 1 to 10.
- In the maintenance planning while conducting a FMECA study on all machines in a plant, machines with a high RPN may undergo predictive maintenance and the machines with a low RPN may undergo reactive maintenance.

Example



- A team of reliability engineers is evaluating four critical machines (M1 to M4) in a process plant using FMECA. For each machine, the rankings for **Occurrence (O)**, **Severity (S)**, and **Detection (D)** are estimated based on historical data and expert judgment. The rankings are as follows:

Machine	Occurrence	Severity	Detection
M1	7	6	4
M2	5	9	6
M3	8	5	7
M4	6	7	5

- $RPN < 150$ is Reactive,
- $150 < RPN < 250$ is Preventive,
- $RPN > 250$ is Predictive

- Calculate the RPN for each machine. Assign a maintenance policy to the machines based on the RPN.
- Suppose you can improve either Occurrence or Detection scores of Machine M2, but not both, due to resource limitations. Which parameter would you choose to reduce to bring the RPN below 250, and what should the new score be?

FMECA Data Collection



Project:

Version:

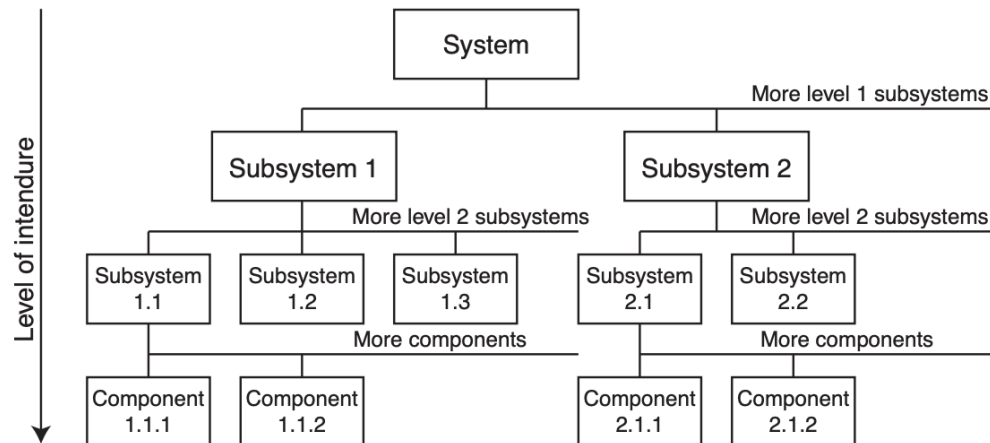
Date:

System:

Subsystem:

Teamwork leader:

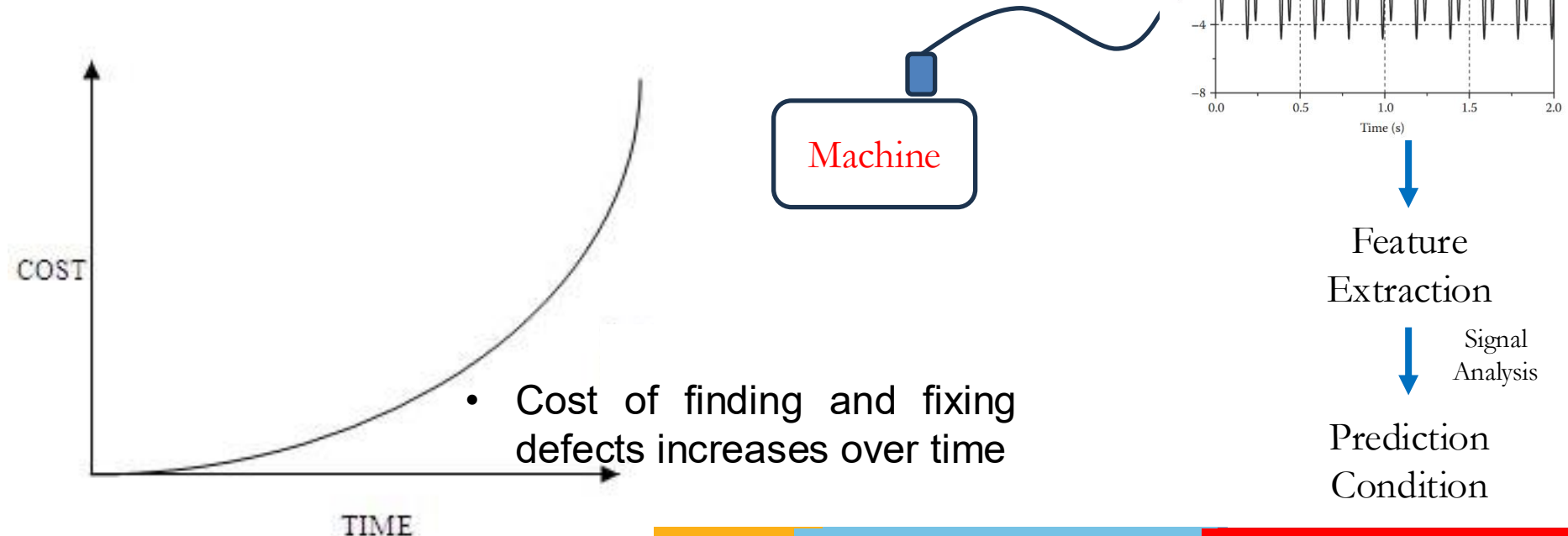
Id.	Comp.	Function	Failure mode	Failure cause	Local effects	Global effects	S	O	D	RPN	Corrective actions



Fault Diagnosis and Prognosis



- Using FMECA one can identify critical machines or components!
- What to do with that information?
- If a fault has occurred!
- What kind of fault has occurred?
- Has it totally failed, or do we still have time?

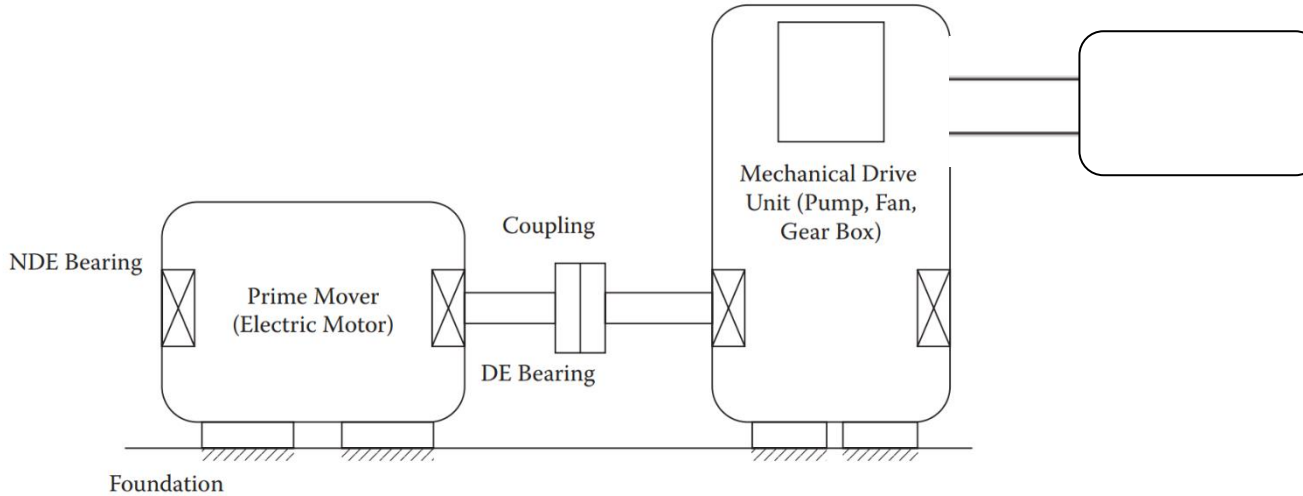


Fault Diagnosis and Prognosis

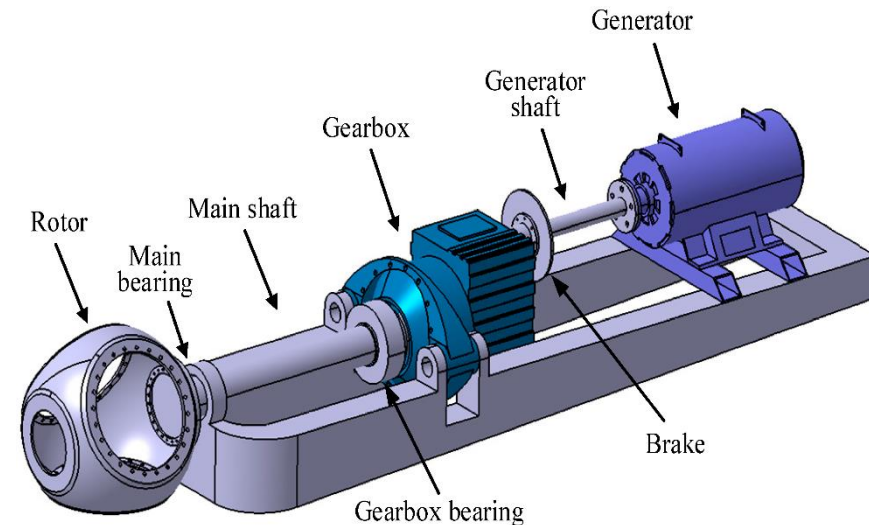


- **Diagnosis:**
- **Detecting, isolating, and identifying** an impending or incipient failure condition.
- The affected component (subsystem, system) is still operational even though at a degraded mode.
- **Prognosis:**
- Ability to predict accurately and precisely the remaining useful life (RUL) of a failing component or subsystem.

Faults in Rotating Machines



- **Shaft misalignment**
- **Shaft unbalance**
- **Shaft crack**
- **Bearing defects**
- **Gear defects**
- **Electrical defects**
- **etc**



Techniques

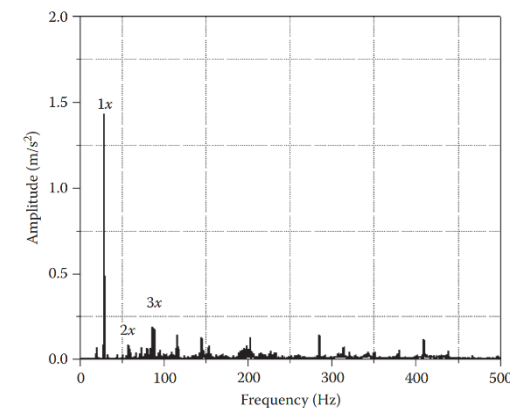
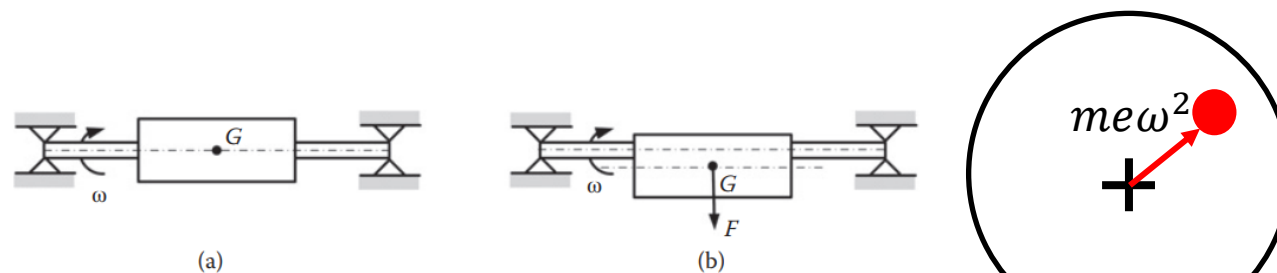


- **Vibration Monitoring**
- **Acoustic Monitoring**
- **Motor Current Signature Analysis**
- **Thermography**
- **Wear Debris Analysis**
- **etc**

Shaft Unbalance

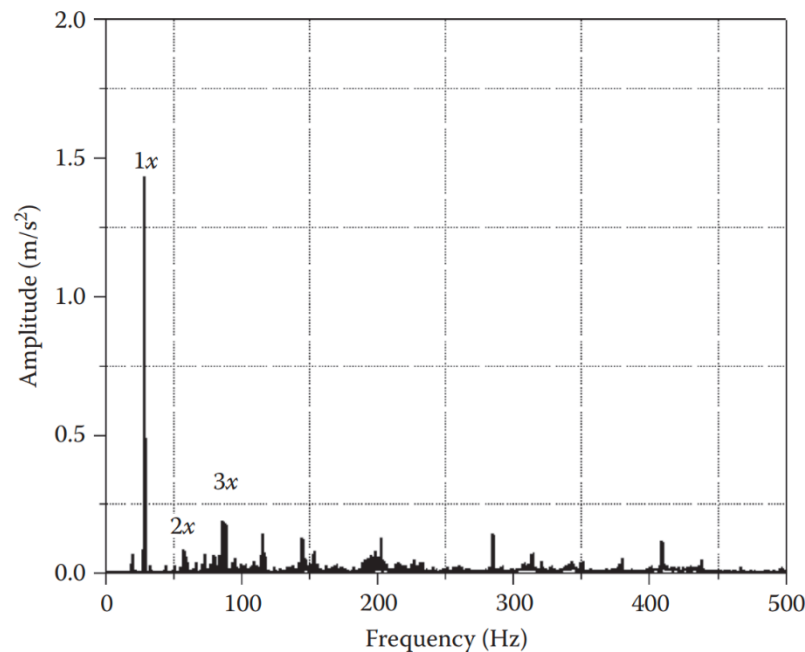


- Unbalance is due to a net uneven distribution of the mass of a rotating component about its axis.
- The unbalance force acts radially and is proportional to the square of the angular rotational speed of the shaft.
- The unbalance force induces additional fatigue load on the bearings and may lead to premature failure of the bearings and rotating shafts.
- Static unbalance and dynamic unbalance



Shaft Unbalance

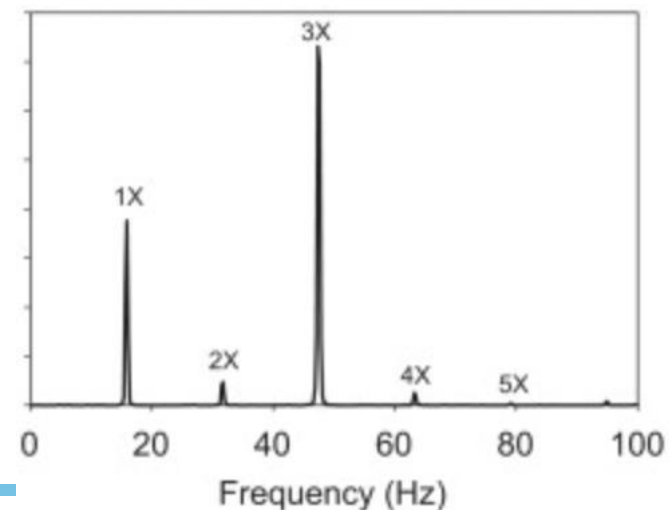
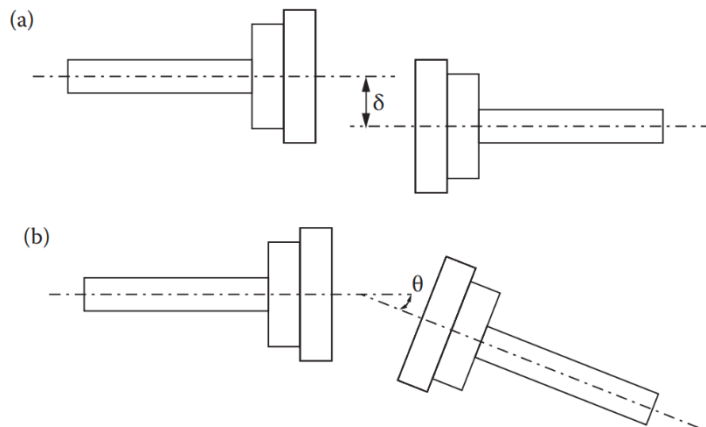
- Vibration Characteristic of Unbalance (rotor between bearings)
- **1X of shaft frequency in Vibration Spectrum**
- Harmonic characteristic
- Magnitude of vibration increases with rotor speed
- Vibration is in-phase between the support bearings



Shaft Misalignment



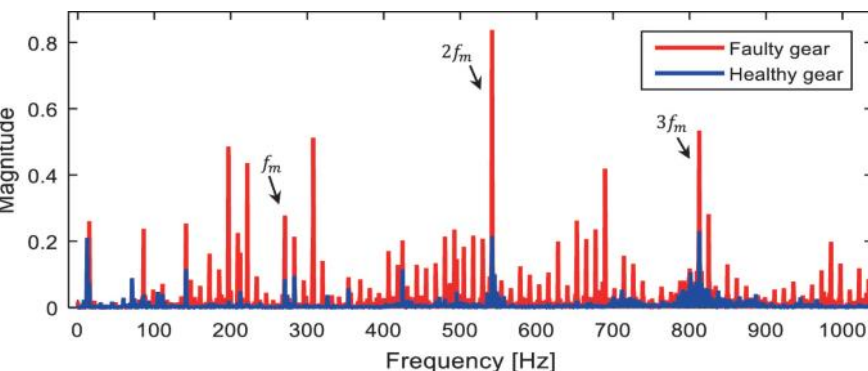
- Due to uneven foundation level, the shafts are not concentric and they are offset either by the same lateral amount or by an angle.
- This condition leads to the misalignment of the shaft, which can be **parallel and angular**.
- Vibration monitoring helps detect the misalignments in shafts.
- It leads to the increase in the axial vibration levels at higher frequencies, like **twice or thrice the rotating speeds of the shaft**.
- This is because an increase in the misalignment can cause the torque on the shaft to increase at higher rotational speed.



Gear Faults



- Gears are responsible for power transmission between two shafts at the desired rotation speeds.
- **During meshing**, the gears are in contact with each other, and as soon as they leave contact due to a sudden decrease in the torque transmission, the gears are set into torsional vibration.
- This is transmitted to the shafts, bearings, and finally the gear box housing or casing.
- By measuring these vibration and their spectra, one can estimate presence of any fault.
- Healthy gear spectrum will have peaks at the **gear mesh frequency (GMF)**.
- Any fault will result in presence of sidebands around the GMF.



Healthy

$$x(t) = \sum_{n=1}^N \sin(2\pi n f_m t + \psi)$$

Faulty,

$$x(t) = \left(1 + \sum_{k=1}^K \sin 2\pi k f_{sh} t\right) \sum_{n=1}^N \sin(2\pi n f_m t + \phi)$$

Gear Faults

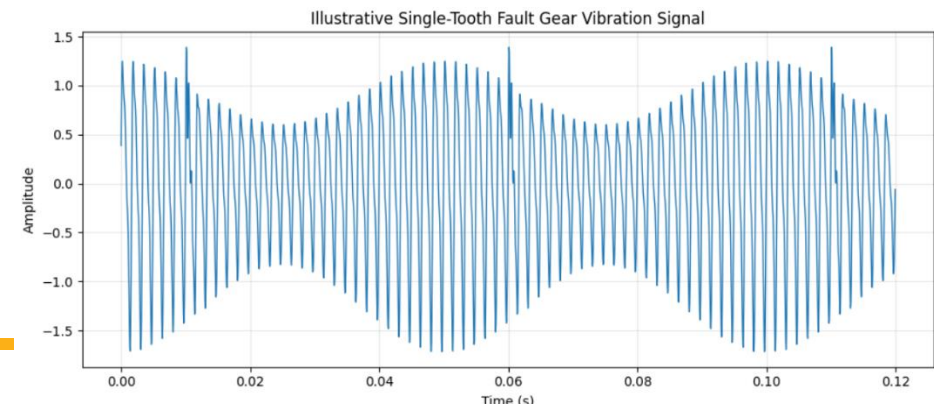
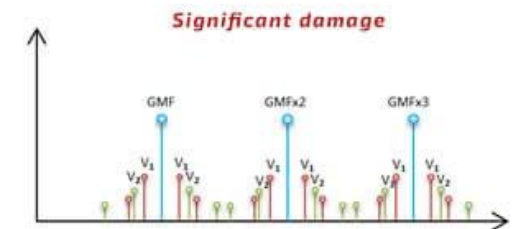
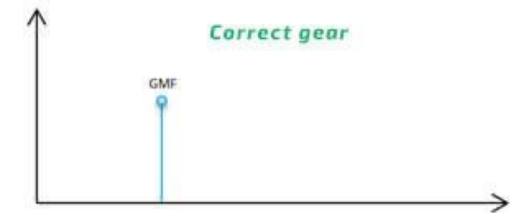
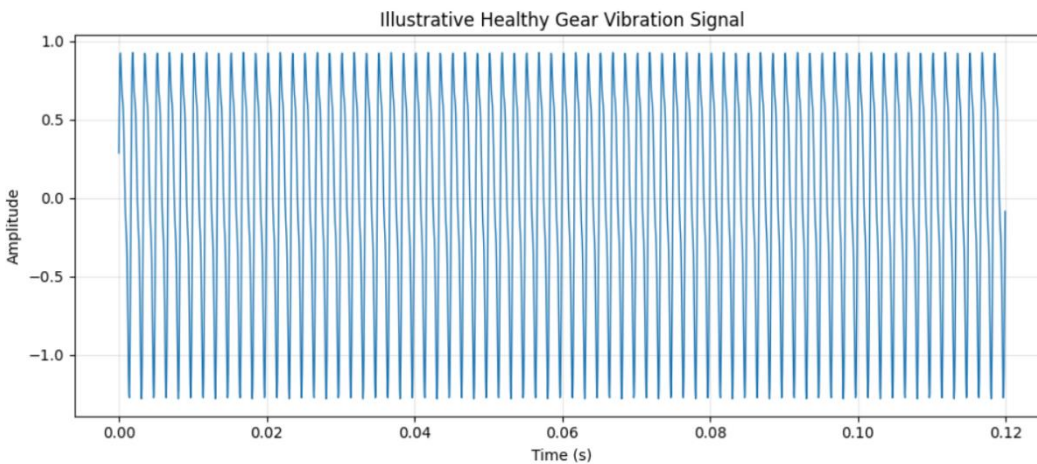
innovate

achieve

lead

$$x(t) = \sin(2\pi n f_m t + \psi)$$

$$x(t) = (1 + \sin 2\pi n f_{sh} t) \sin(2\pi n f_m t + \phi)$$

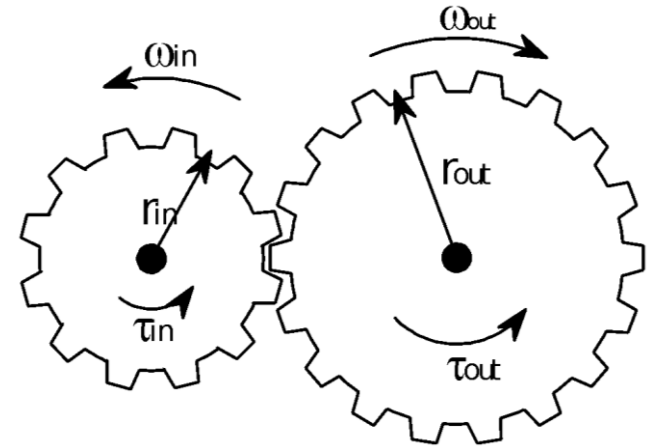


Gear Characteristic Frequencies

- For fixed-axis gear-sets**

- Shaft frequency, f_{sh}
- Gear-meshing frequency, $f_m = N_t * f_{sh}$
- Fault characteristic frequency, $f_{ch} = f_m \pm f_{sh}$

$$GMF[Hz] = \frac{N_G T_G}{60} = \frac{N_P T_P}{60}$$

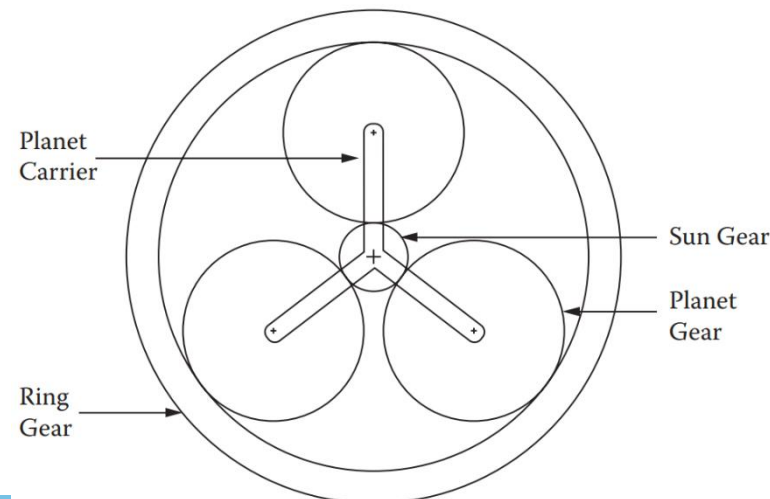


- For Planetary gears**

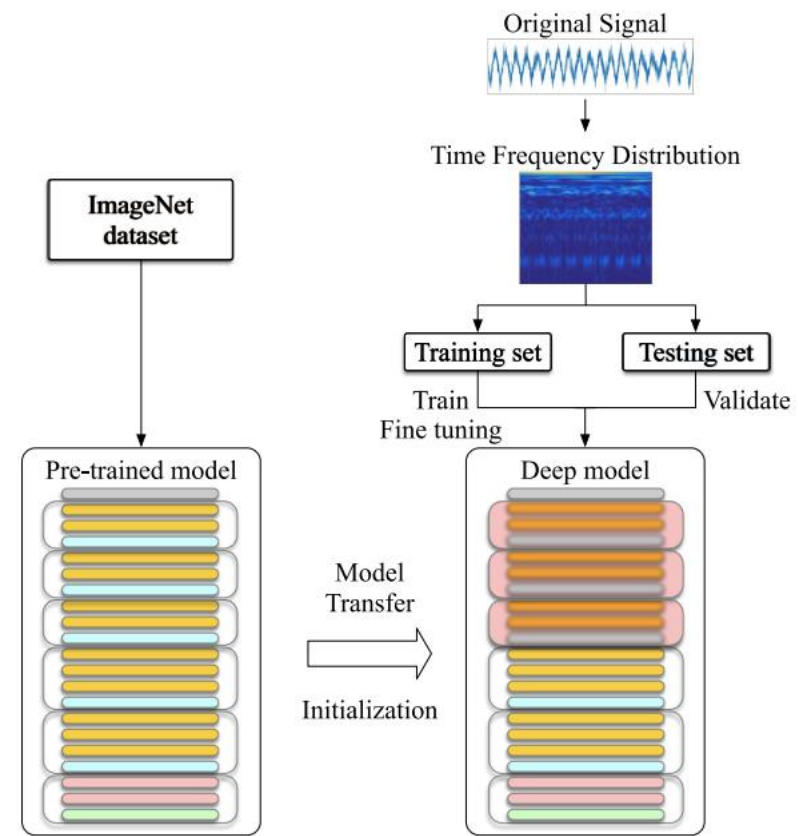
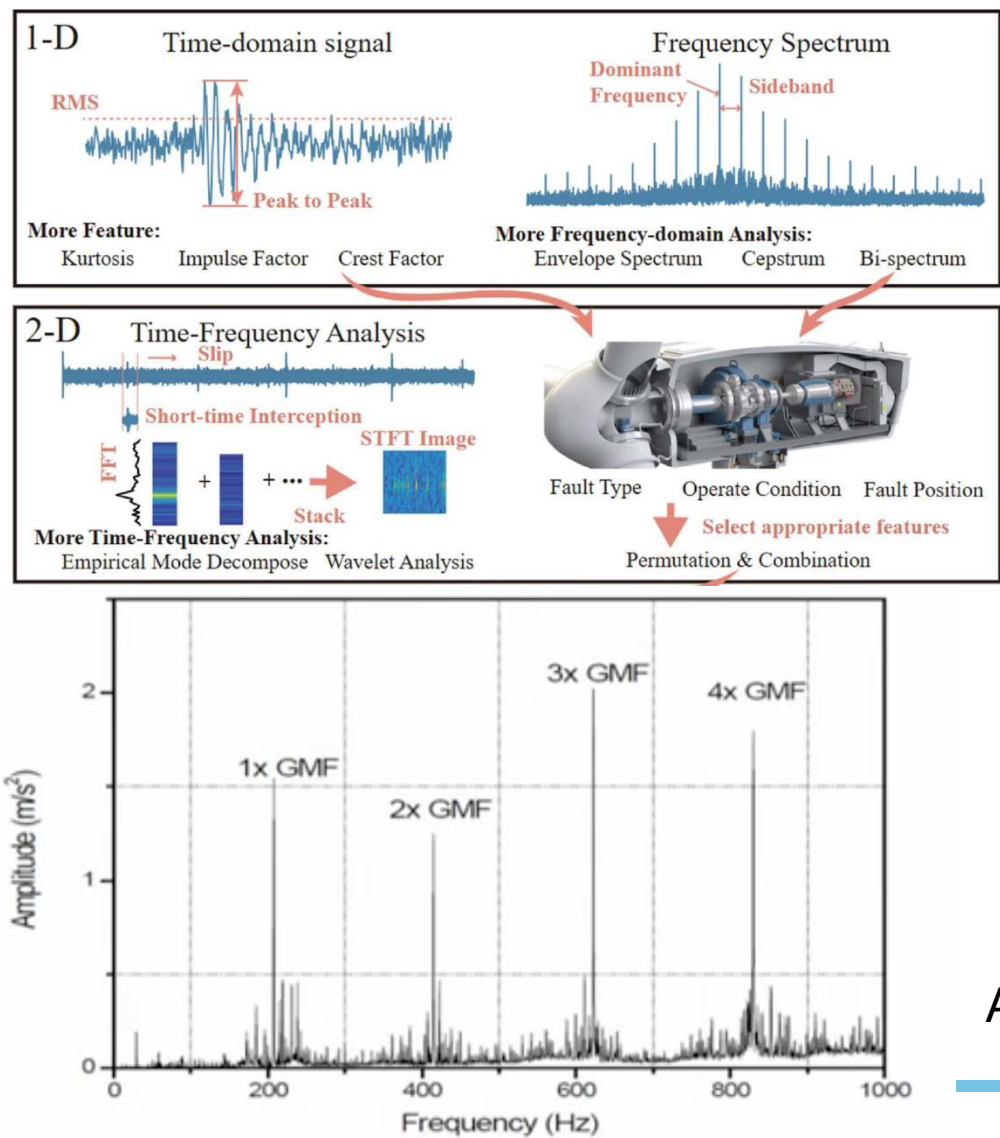
- Two types of GMFs are present.
- GMF between planet carrier and ring
- GMF between sun and planet carrier

$$GM_{s/p} = |(N_s - N_c)| T_s$$

$$GM_{r/p} = |(N_r - N_c)| T_r$$



Gear Characteristic Frequencies



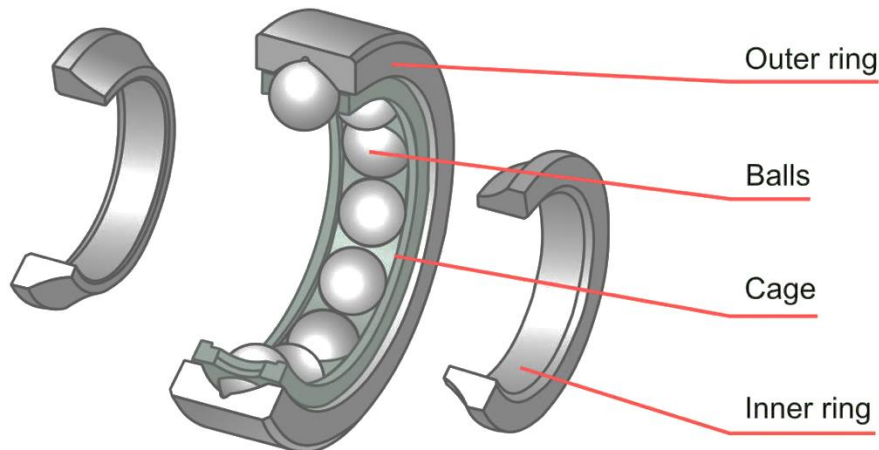
AlexNet, ResNet etc

Time Domain Analysis

- Time domain statistical features like mean, STD, RMS, Kurtosis, Crest Factor etc also can be adopted for analysing the health of a gear.
 - For more details read:
1. **Sharma, Vikas, and Anand Parey. "A review of gear fault diagnosis using various condition indicators." *Procedia Engineering* 144 (2016): 253-263.**
 2. **Zhu, Junda, et al. "Survey of condition indicators for condition monitoring systems." *Annual Conference of the PHM Society*. Vol. 6. No. 1. 2014.**

Bearing Faults

- Bearings are an important machine element that supports the rotor system.
- In machineries, the two most common types of bearings are used—the antifriction bearing or the **rolling element bearing**, and the **journal bearing** working on the principle of hydrodynamic lubrication.
- We will focus on Rolling Element Bearings (REBs).
- Rolling element bearings have **four very important components**.
- Inner race, Outer race, Rolling element (balls/rollers), and Cage

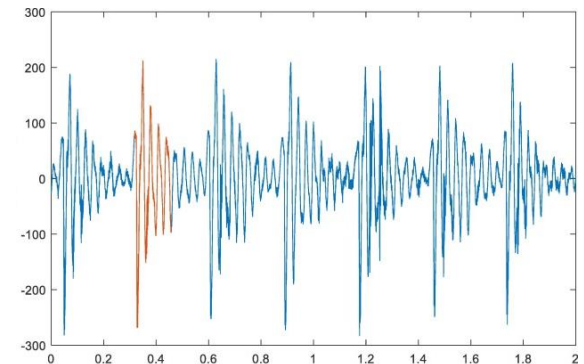


Bearing Vibration Signal

- Bearing vibration signal can be modelled as a system of impulses modulated by the defect frequency.
- A defect in a bearing excites the **system resonance frequency f_n** which acts as the **carrier frequency** and it gets **modulated by the bearing defect characteristic frequency**.

$$x(t) = \sum_{i=-\infty}^{i=+\infty} A(iT) s(t - iT - \tau)$$

$$x(t) = \sum_k A_k e^{-\zeta \omega_n(t - kT_f)} \sin(\omega_d(t - kT_f)) u(t - kT_f)$$



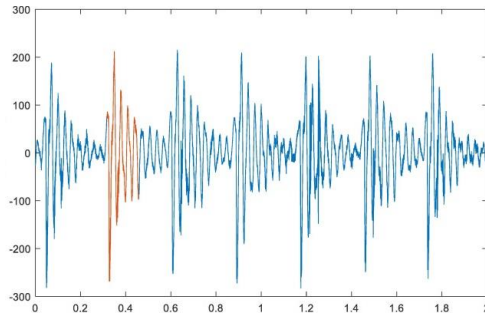
$$x_{OR}(t) = \sum_{i=-\infty}^{+\infty} A(iT_{BPFO}) e^{-\zeta \omega_n(t - iT_{BPFO} - \tau)} \sin[\omega_d(t - iT_{BPFO} - \tau)] u(t - iT_{BPFO} - \tau)$$

$$x_{IR}(t) = \sum_{i=-\infty}^{+\infty} A_0 [1 + \alpha \cos(2\pi f_r iT_{BPFI} + \phi)] e^{-\zeta \omega_n(t - iT_{BPFI} - \tau)} \sin(\omega_d(t - iT_{BPFI} - \tau)) u(t - iT_{BPFI} - \tau)$$

Bearing Characteristic Frequencies



Any defect in each of the four components, gives rise to a unique characteristic frequency.



$$x(t) = \sum_{i=-\infty}^{i=+\infty} A(iT) s(t - iT - \tau)$$

Ballpass frequency, outer race:

$$BPFO = \frac{nf_r}{2} \left\{ 1 - \frac{d}{D} \cos \phi \right\}$$

Ballpass frequency, inner race:

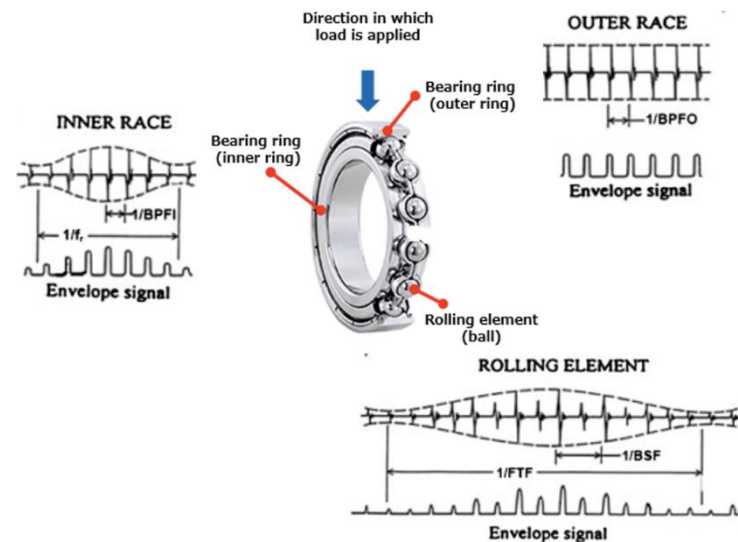
$$BPFI = \frac{nf_r}{2} \left\{ 1 + \frac{d}{D} \cos \phi \right\}$$

Fundamental train frequency (cage speed):

$$FTF = \frac{f_r}{2} \left\{ 1 - \frac{d}{D} \cos \phi \right\}$$

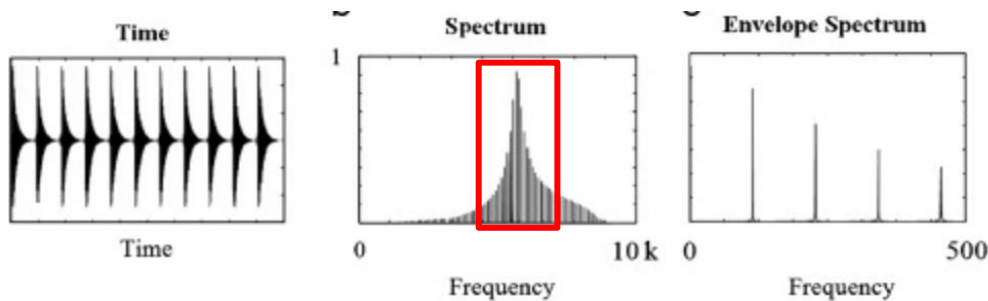
Ball (roller) spin frequency:

$$BSF(RSF) = \frac{D}{2d} \left\{ 1 - \left(\frac{d}{D} \cos \phi \right)^2 \right\}$$



Filtering around Resonance

For noisy signals you may filter around the resonance band to get a signal with better SNR.



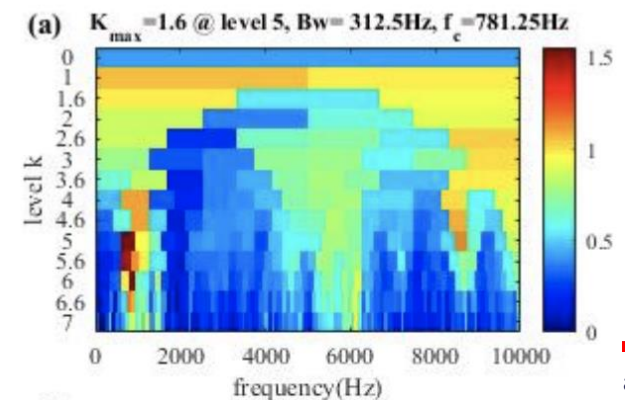
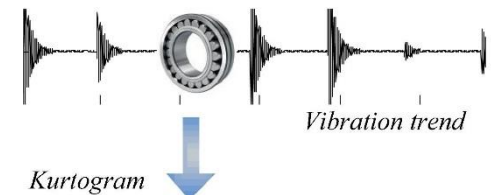
Resonance Frequency band

- Estimate the resonance band and design a band-pass filter
- Band may be estimated by analysing the raw signal spectrum

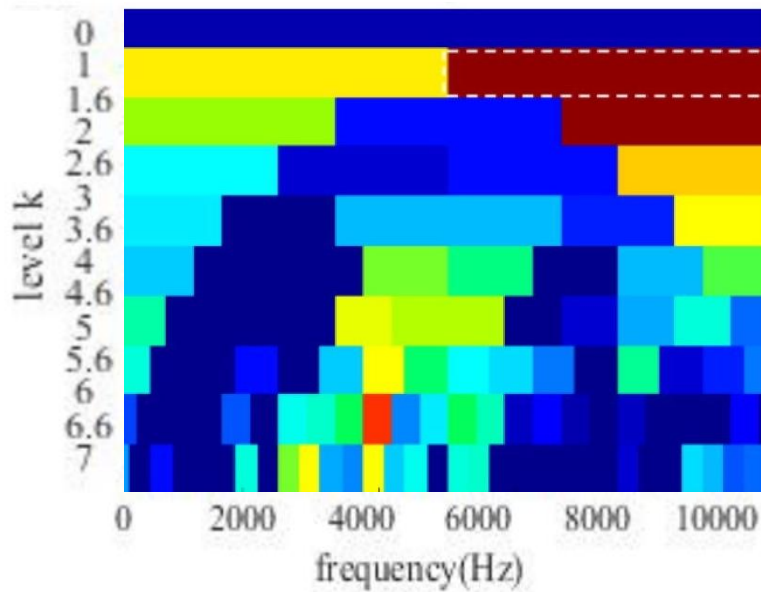
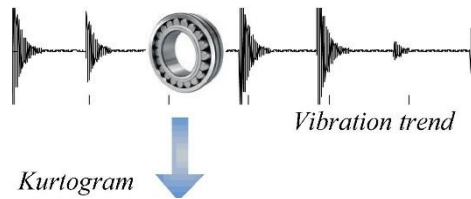
- Spectral Kurtosis (SK) based analysis is also adopted to find the resonance band

$$SK(f) = \frac{\langle |X^4(t, f)| \rangle}{\langle X^4(t, f) \rangle^2} - 2$$

- The SK, of a signal, is defined as the Kurtosis of its spectral components.



Kurtogram

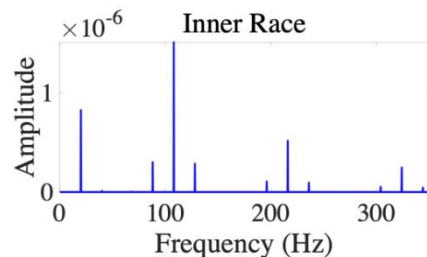
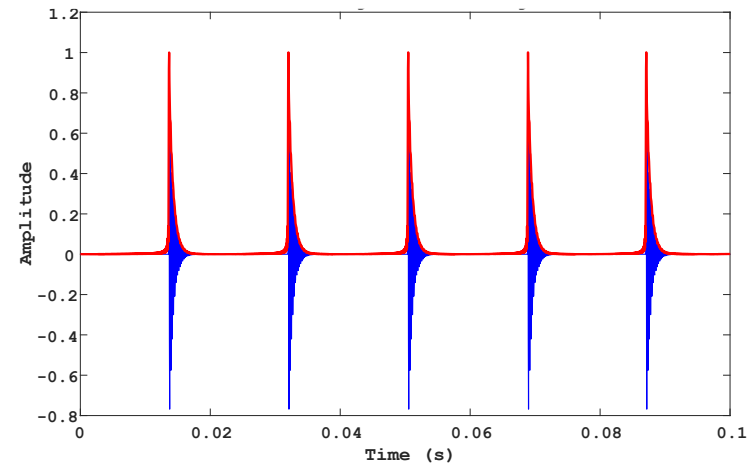
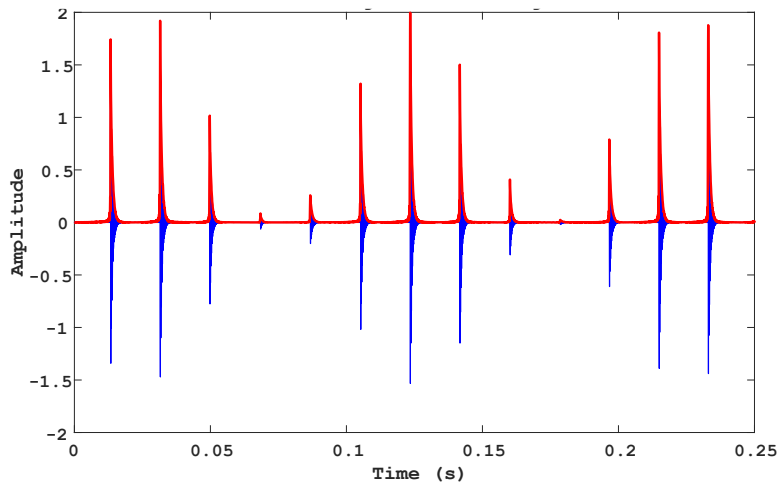


$$SK(f) = \frac{\langle |X^4(t,f)| \rangle}{\langle X^4(t,f) \rangle^2} - 2$$

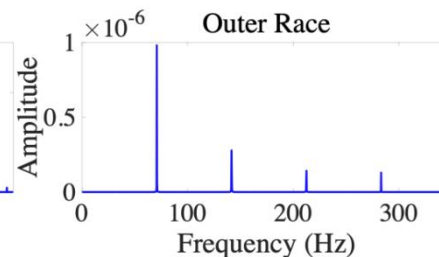
$X(t, f)$ is the STFT of the vibration signal

Envelop Analysis

- For bearings, it is observed that the bearing **envelop signal spectrum** contains the fault specific information.
- Therefore, envelop signal needs to be extracted from the raw signal by demodulation



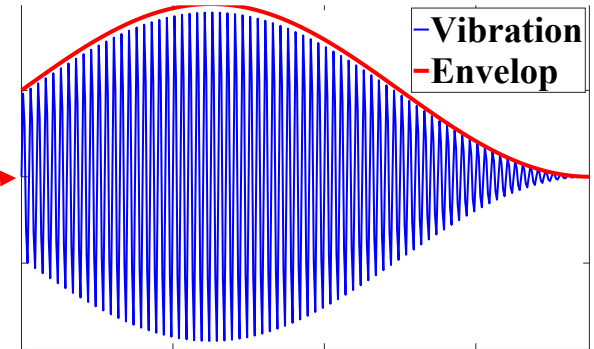
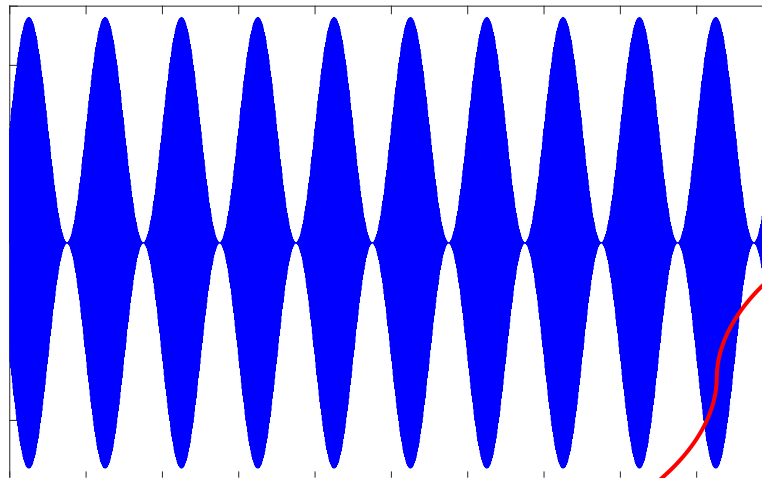
(a)



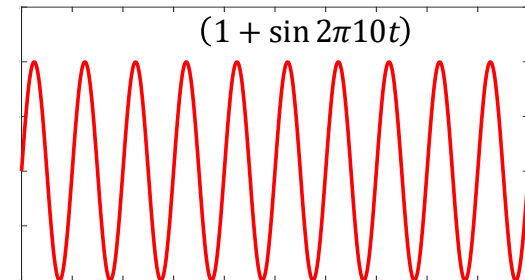
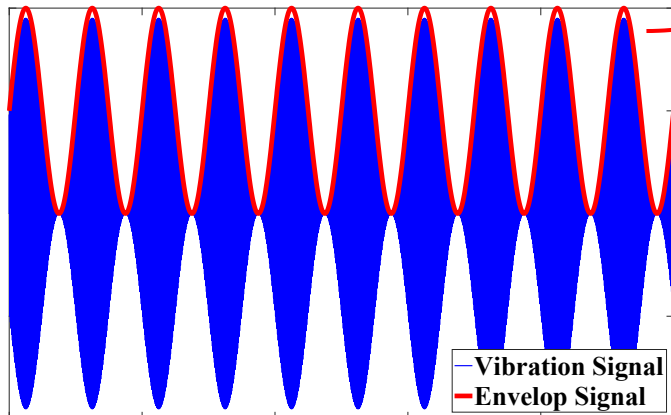
(b)

Envelop Signal Basics

$$x(t) = (1 + \sin 2\pi 10t) \sin(2\pi 1000t)$$

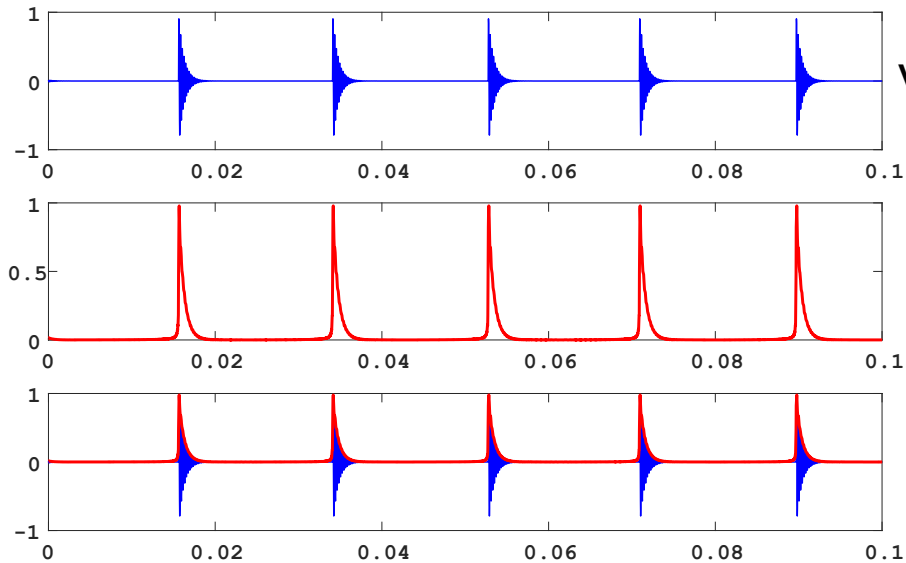


Hilbert Transform based Demodulation gives the Envelop Signal



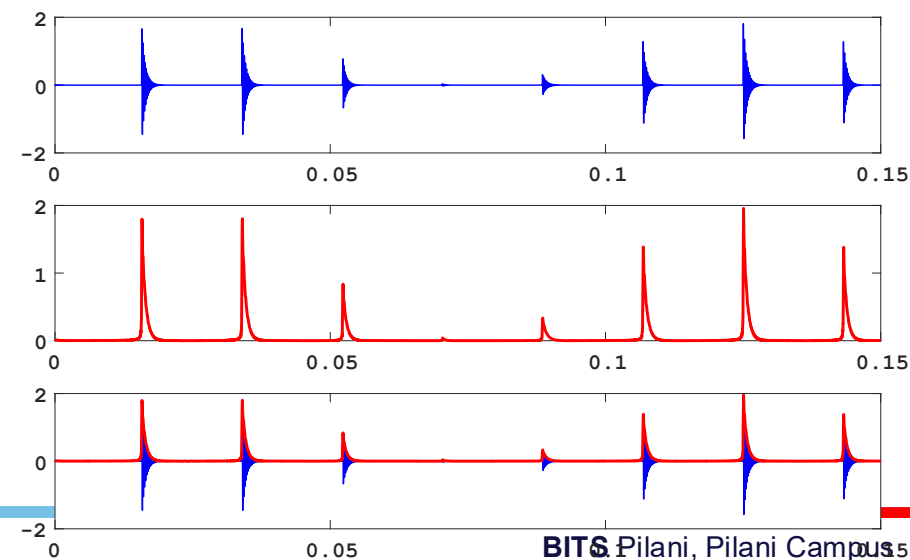
Envelop signal = abs(Hilbert(time domain signal));

Bearing Envelop Signals



Outer Race

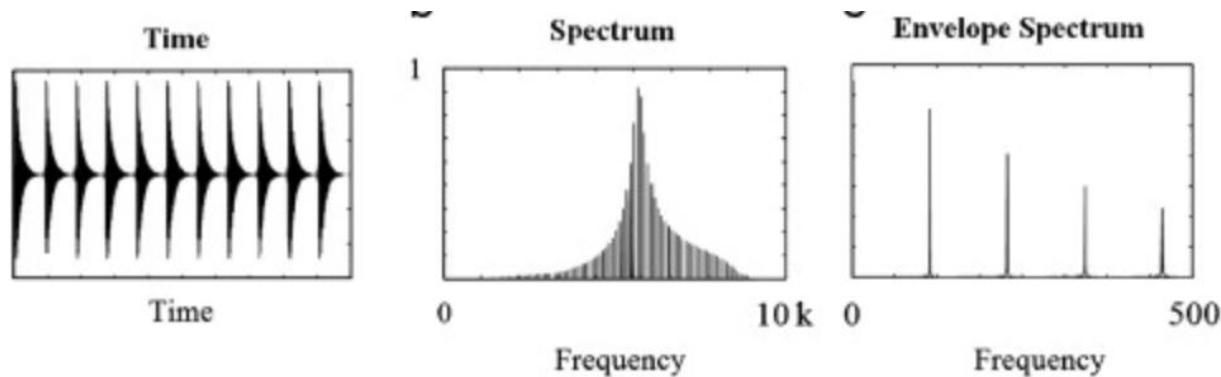
Inner Race



Envelop Spectrum

To obtain the envelop spectrum, the steps are:

1. Bearing vibration signal **Filtering around resonance frequency**
2. Demodulate vibration signal using Hilbert Transform **Envelop signal = $\text{abs}(\text{Hilbert}(\text{time domain signal}))$;**
3. Take FFT of the Demodulated signal

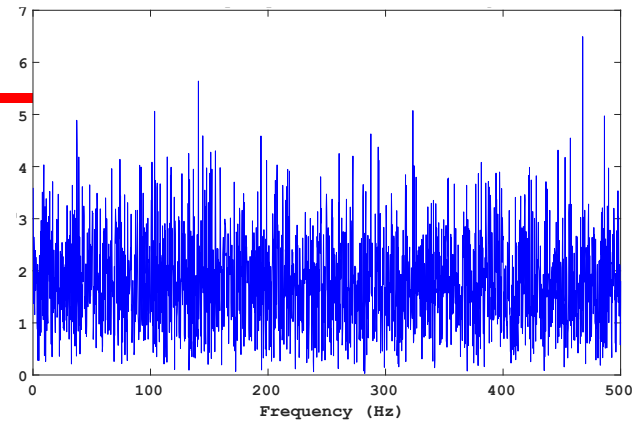
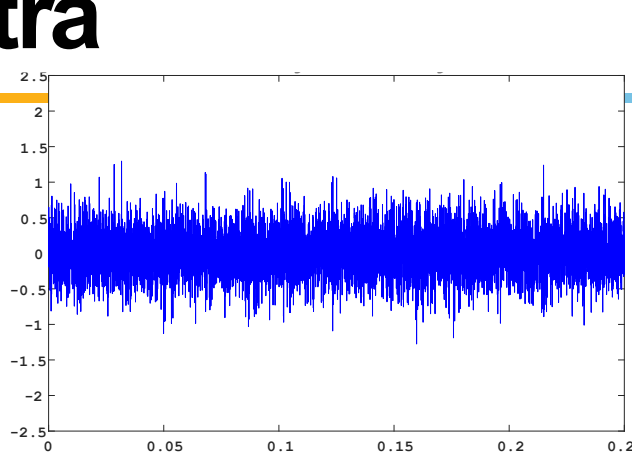


With increase in fault severity the amplitude of the fault frequencies should ideally rise.

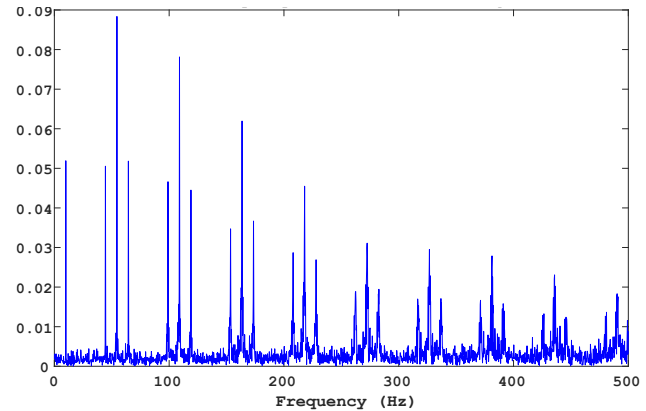
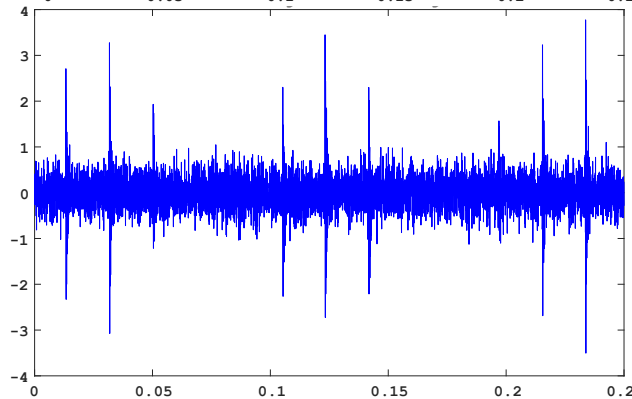
Bearing Signals and Envelop Spectra



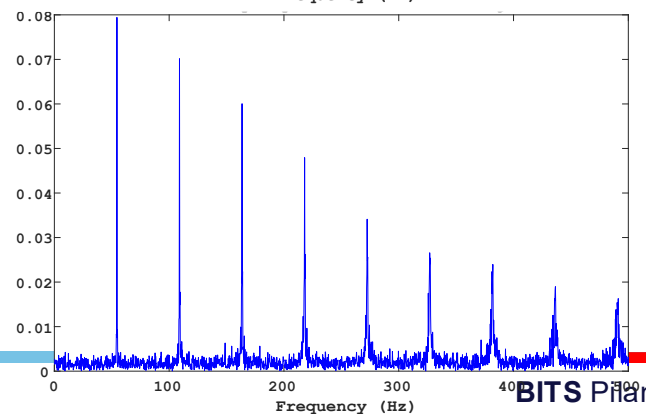
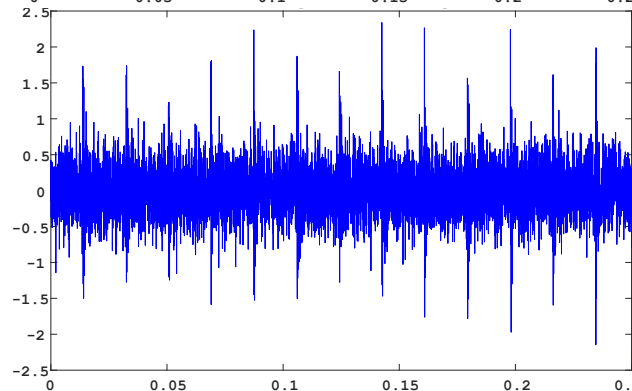
- Healthy



- IR



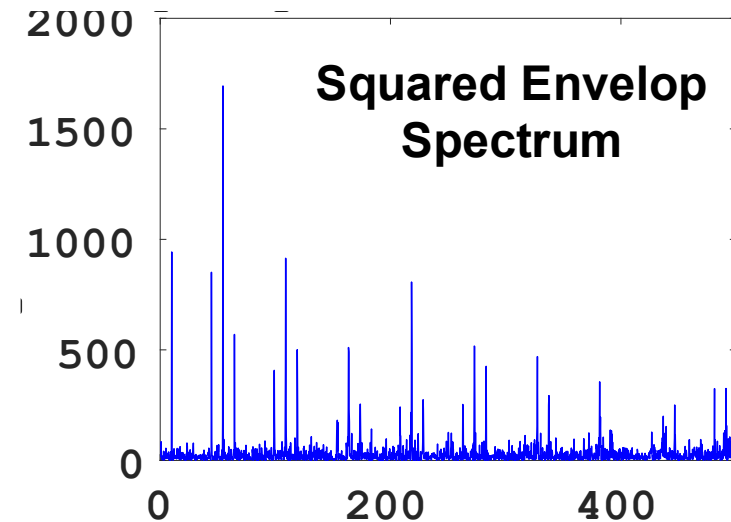
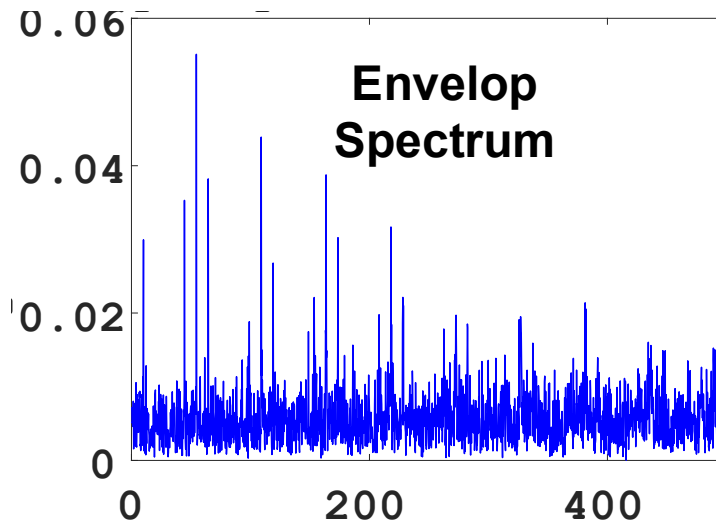
- OR



Squared Envelop Signal Spectrum

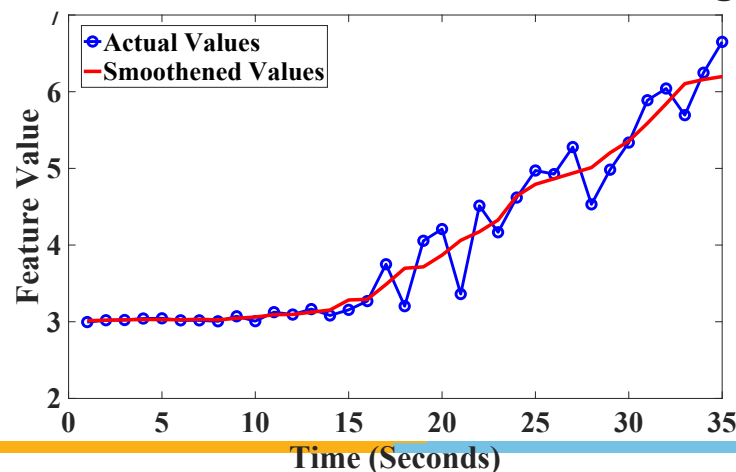


- Bearing defects exhibit cyclostationary characteristics, hence **squared envelop signal** has been found to have better SNR than the normal envelop signal.
- The squared envelop spectrum can be estimated as
- $SES_x = |DFT\{x_{env}^2[n]\}|^2$



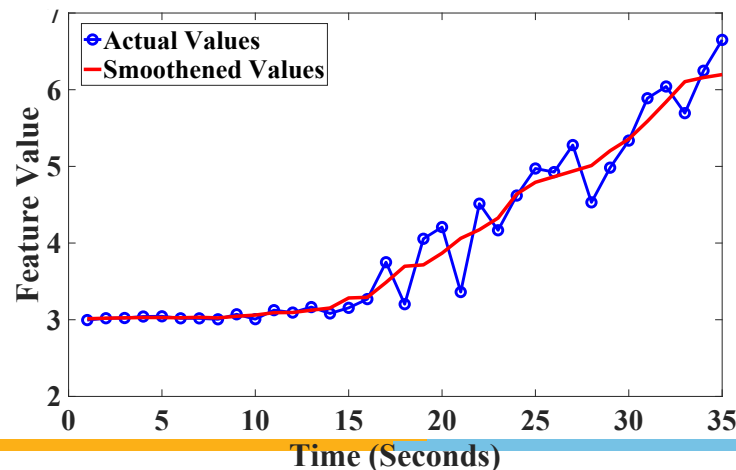
Time Domain Analysis

- Time domain statistical features like mean, STD, RMS, Kurtosis, Crest Factor etc also can be adopted for analysing the health of a bearing.
- "Mean"; "Std"; "Skewness"; "Kurtosis"; "Peak2Peak"; "RMS"; "CrestFactor"; "ShapeFactor"; "ImpulseFactor"; "MarginFactor"; "Energy"
- **These features can be calculated for the raw vibration signal or the envelop/squared envelop signals.**
- The feature values may have randomness because of presence of noise etc.
- In that case filtering or smoothing using moving mean may be adopted.
- This smoothed feature value can be considered as a **Bearing Health Indicator**.

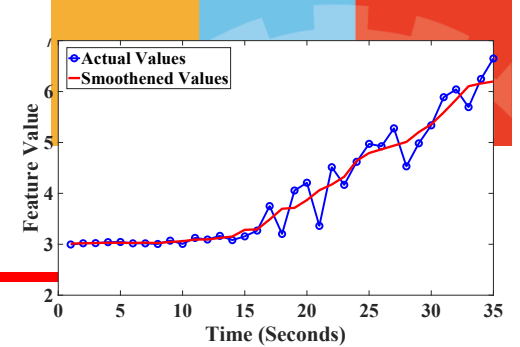


Time Domain Analysis

- "Mean"; "Std"; "Skewness"; "Kurtosis"; "Peak2Peak"; "RMS"; "CrestFactor"; "ShapeFactor"; "ImpulseFactor"; "MarginFactor"; "Energy"
 - To understand what are some good statistical measures for fault monitoring, please refer:
1. Ali, Jaouher Ben, et al. "Online automatic diagnosis of wind turbine bearings progressive degradations under real experimental conditions based on unsupervised machine learning." *Applied Acoustics* 132 (2018): 167-181.
 2. Zhu, Junda, et al. "Survey of condition indicators for condition monitoring systems." *Annual Conference of the PHM Society*. Vol. 6. No. 1. 2014.



Health Indicator



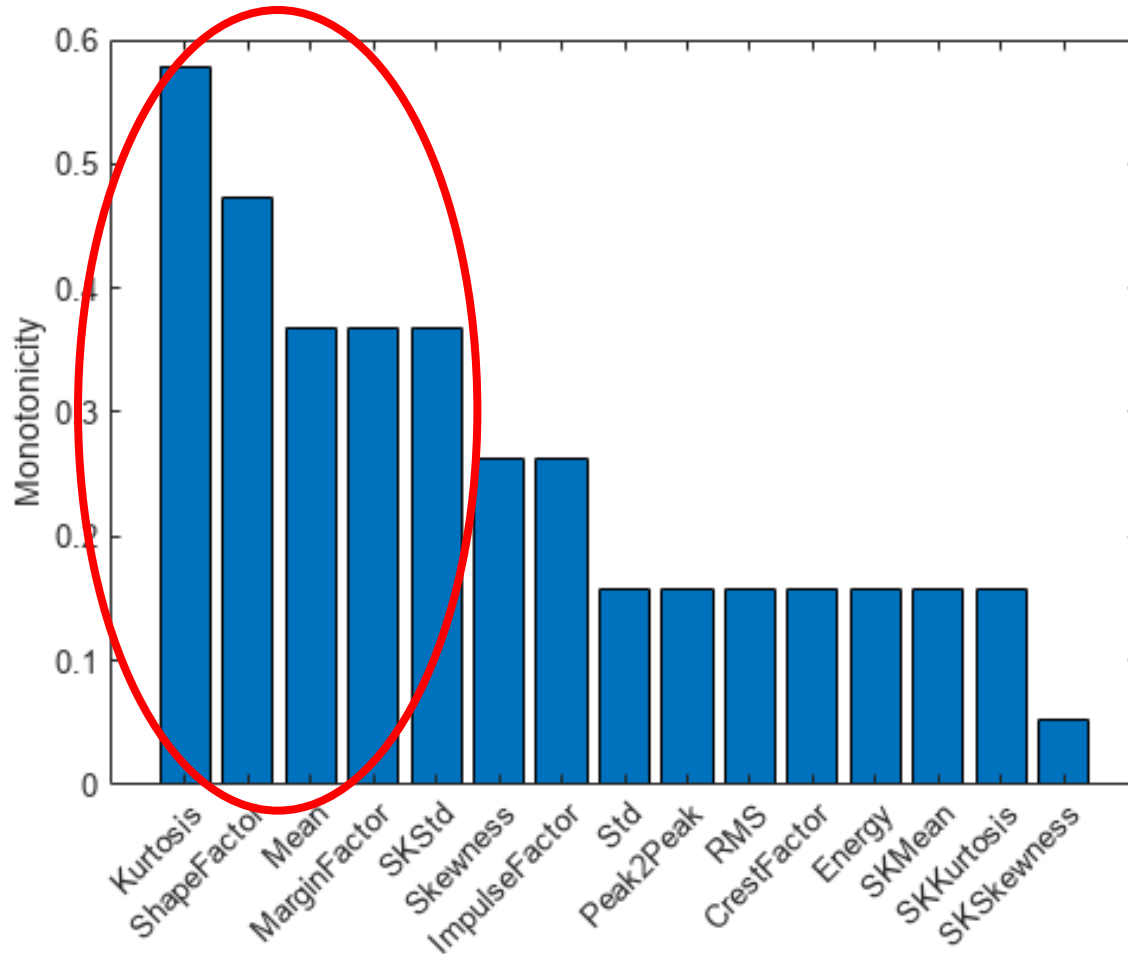
- The **feature value** in the previous slide is the **Health Indicator**.
- **Monotonicity** is an important characteristics in designing a Health Indicator.
- We need an indicator that is always increasing (or decreasing).
- If we have several features (10, 20 or even 30).
- How do we make sure if it is monotonic?
- If we are unable to find a single feature that is monotonic then we may try to estimate the monotonicity of a feature as

$$M(x_i) = \frac{1}{m} \sum_{j=1}^m \left| \frac{\text{number of positive diff}(x_i^j) - \text{number of negative diff}(x_i^j)}{n - 1} \right|$$

where n is the number of measurement points and m is the number of machines monitored.

- Here $M(x_i)$ is the monotonicity of the i -th feature x_i
- Higher the value of $M(x_i)$, better it is.
- Here $\text{diff}(x) = x(t) - x(t - 1)$
- It calculates differences between adjacent elements of any array X .

Health Indicator

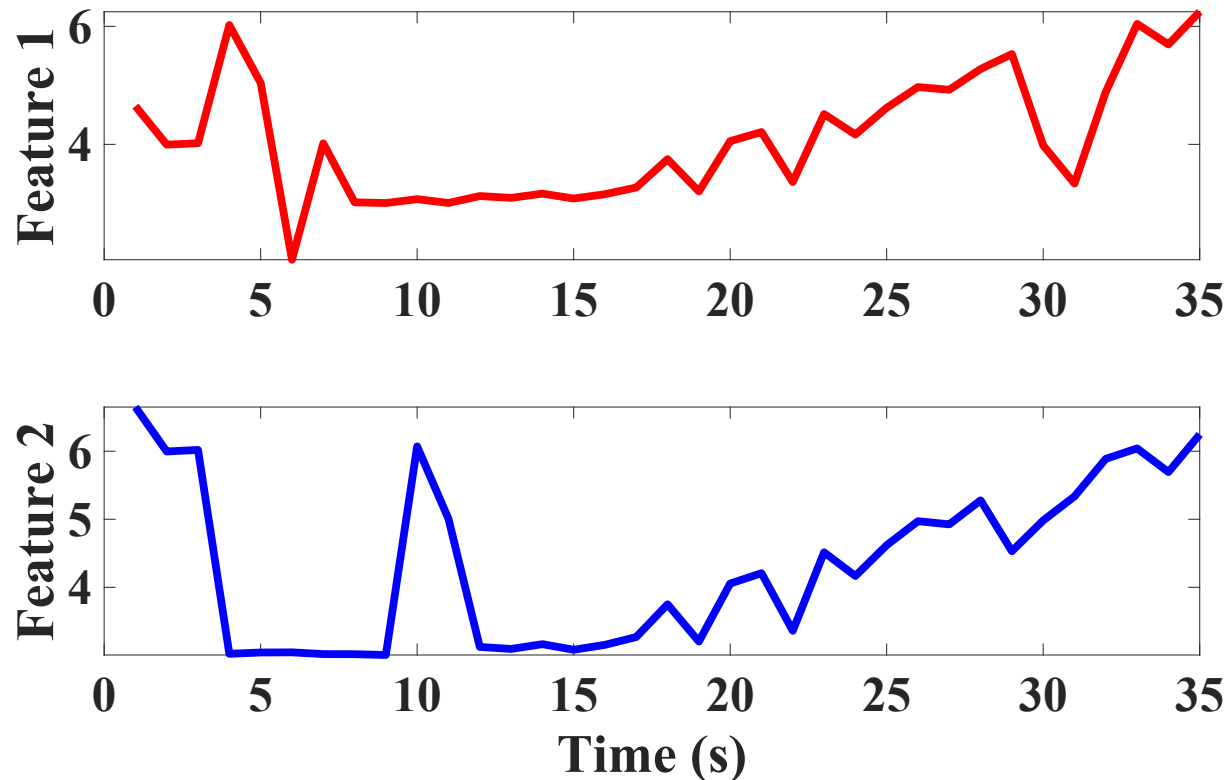


- Let's say we have 15 features.
- We calculate the monotonicity of each feature
- Plot it
- If the feature with highest M shows good monotonic behaviour, use it as an indicator.
- Else select multiple features that have monotonicity **above a certain threshold**.
- **Build a health indicator fusing them.**
- Higher the monotonicity better is the indicator.

Multiple Features

$featureMatrix = [F1, F2, F3, F4, F5]$

- You may get multiple features that are somewhat monotonic but cannot be used as an indicator individually



Feature Fusion

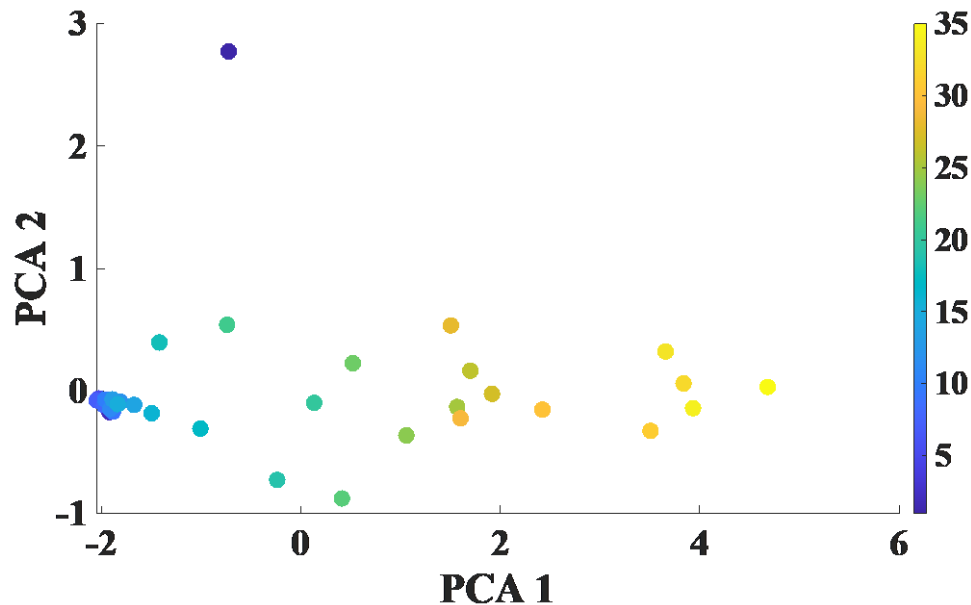
featureMatrix = [*F1*, *F2*, *F3*, *F4*, *F5*]

- If you get multiple features that are monotonic, you may fuse them to design a better Health Indicator.
- Use of dimensionality reduction techniques may be adopted.
- Principal Component Analysis (PCA) etc.
- Let us take an example
- *featureMatrix* = [*RMS*, *Kurtosis*, *Mean*, *STD*, *CrestFactor*]
- Normalize the features first.
- *featureNormalized* = (*featureMatrix* - *meanFeature*) ./ *stdFeature*;
- Reduce the normalized features by PCA.
- [*coeff*, *score*, ~] = *pca*(*featureNormalized*); (MATLAB function)
- Find PCA1 & PCA2

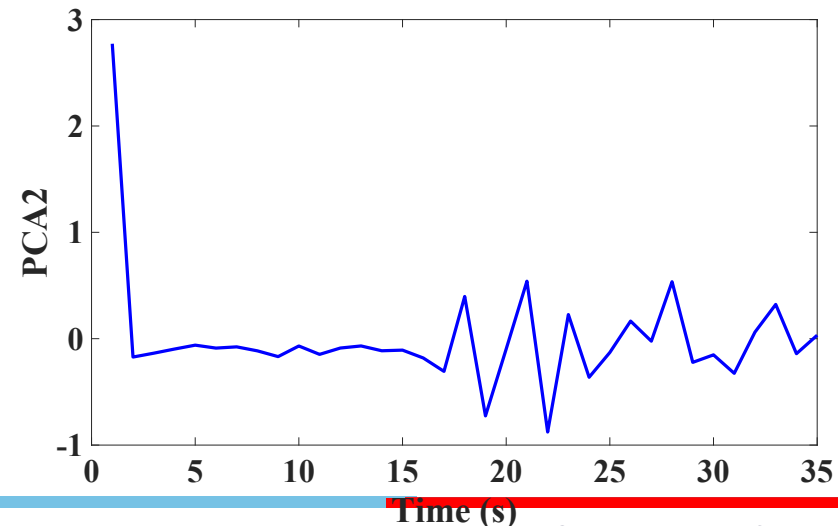
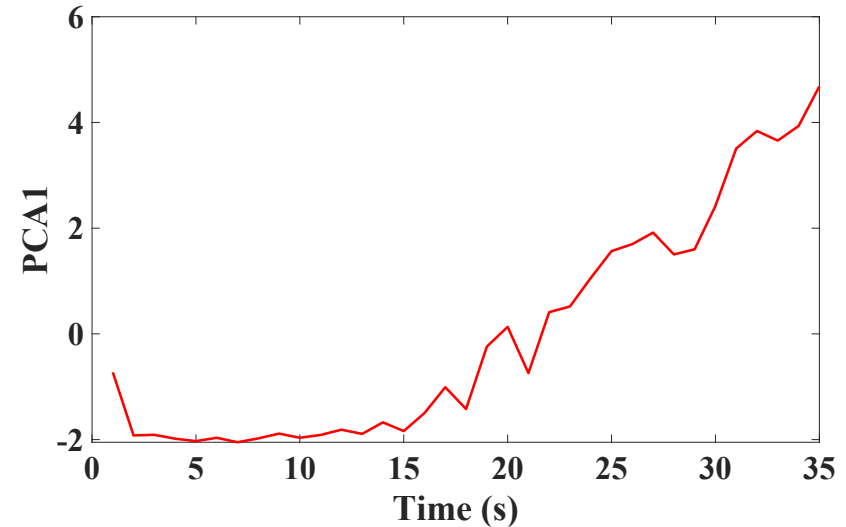

```
PCA1 = (featureSelected{:, :} - meanTrain) ./ sdTrain * coef(:, 1);
PCA2 = (featureSelected{:, :} - meanTrain) ./ sdTrain * coef(:, 2);
```
- Plot both and see which has a monotonic behaviour.

Feature Fusion

- featureMatrix = [RMS, Kurtosis, Mean, STD, CrestFactor]

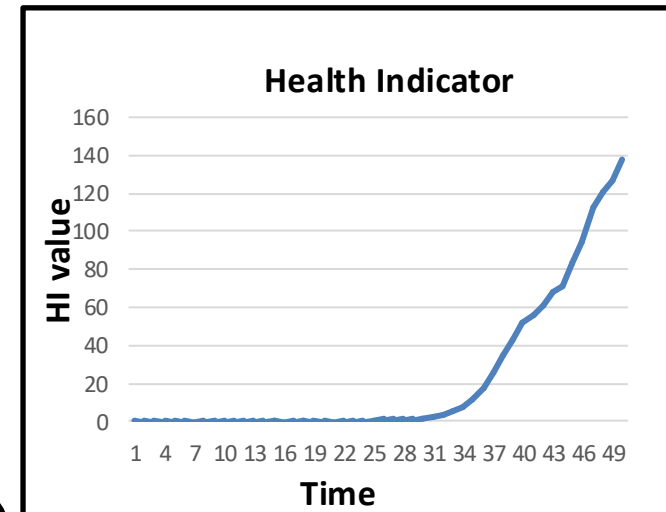
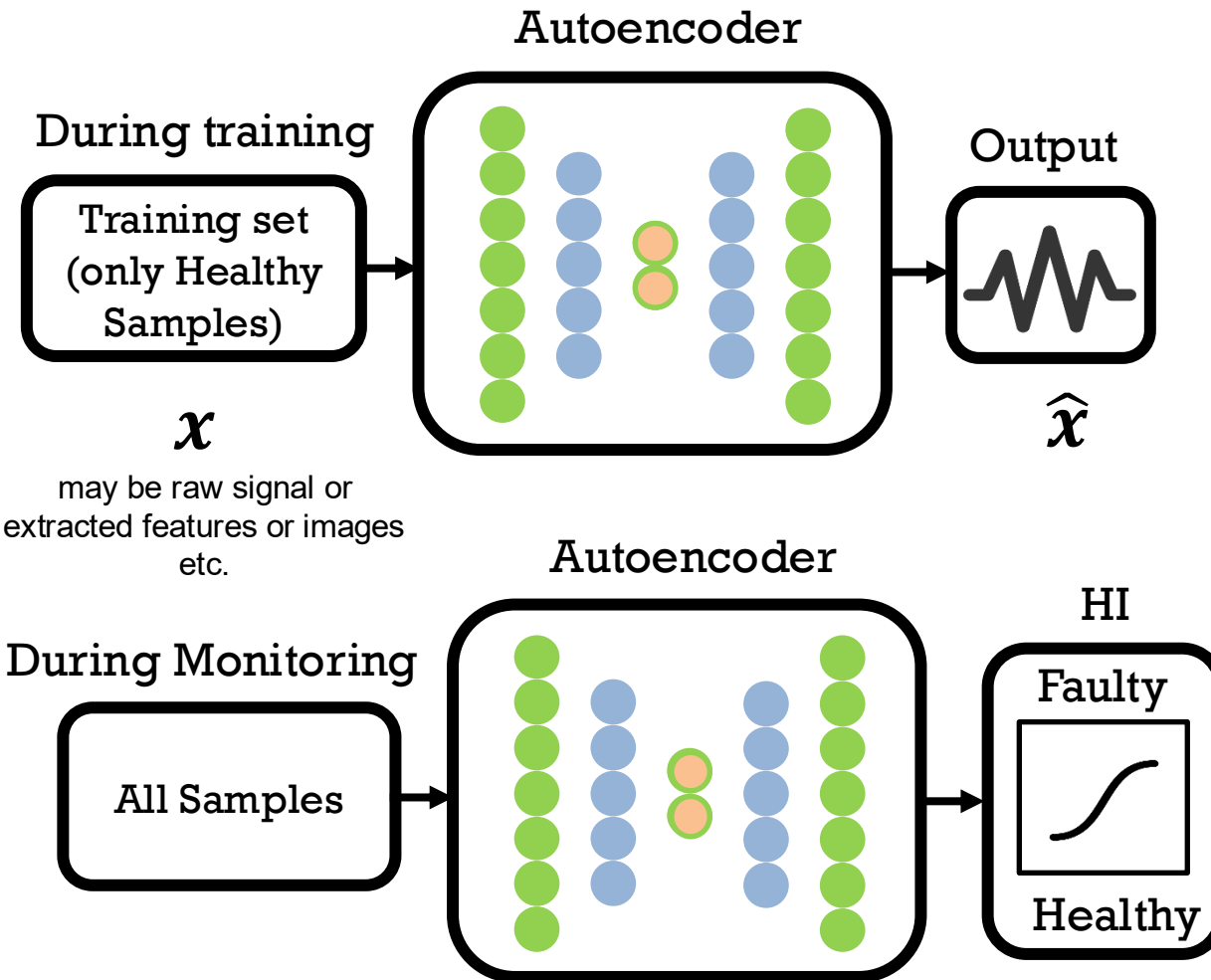


- Here PCA1 seems to be a good Health Indicator



Time (s)

Health Indicator using Deep Learning



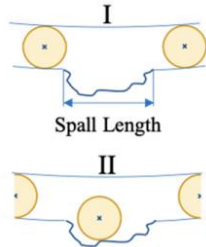
$$\mathbf{HI} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2$$

During monitoring, here x_i is any input sample and $\hat{x}_i$ is the reconstructed sample by the trained autoencoder

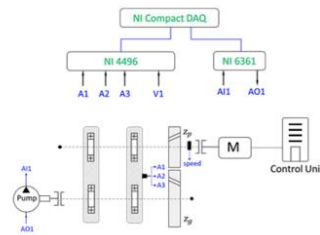
Health Indicators: Physics Based



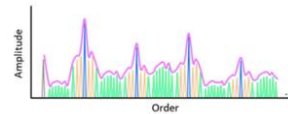
Studying the **physical phenomena**



Vibration **data collection**



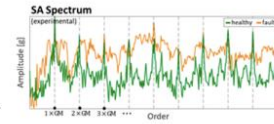
Signal Processing



The **SA signal** can be decomposed into three signals



Physical feature extraction



$$\text{Spectral Energy} = \sum_{i=1}^{N_{ord}} |X(ord_i)|$$



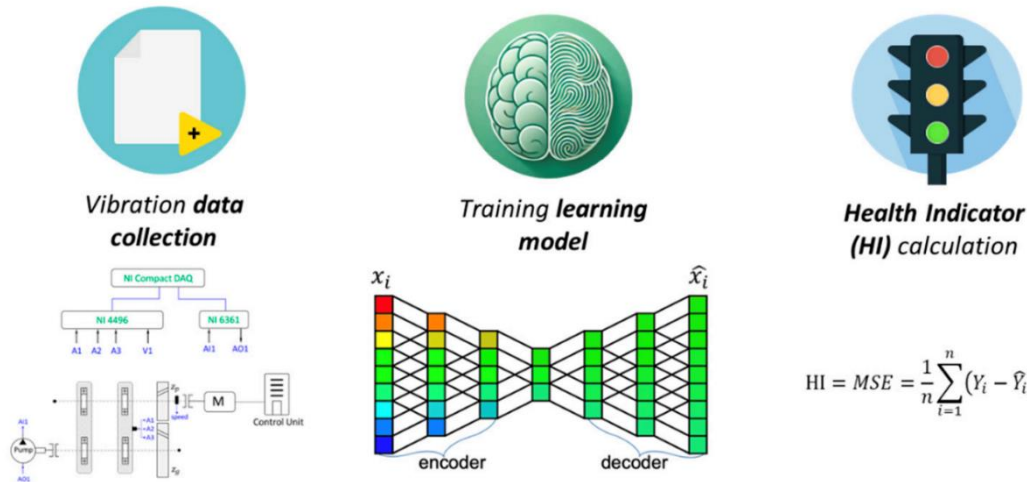
Health Indicator (HI) calculation

$$HI = \sqrt{\frac{1}{M-1} \sum_{m=1}^M \left(\frac{CI_m - \mu_{hm}}{\sigma_{hm}} \right)^2}$$

HI = Root mean square of standardized CI's (Z-score from the healthy case)

- Physics-based approaches **utilize signal processing techniques** designed to **highlight the physical behavior** of faults in vibration signals and to extract meaningful features that can be combined into HIs for monitoring the health status of components.
- Key **advantage include interpretability**, as operators can understand the physical causes of damage based on the model's output.
- However, the models rely on broad assumptions to balance simplicity and realism, making **them less applicable to complex systems**.

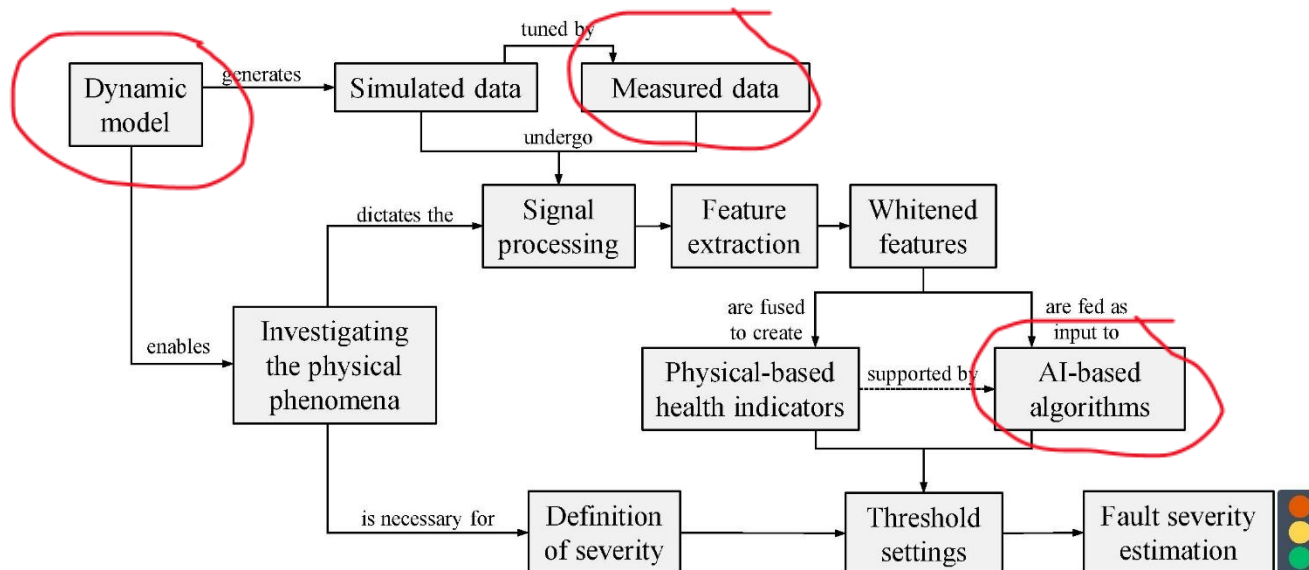
Health Indicators: Data Driven



- Data-driven approach **leverages ML, and AI** to diagnose faults by identifying patterns in large datasets without relying on specific system assumptions.
- It typically involves **learning a complex function that transforms** raw data or extracted features **into a scalar HI**, often using supervised or unsupervised learning models.
- It is particularly **effective for complex systems where deriving an accurate physical model is challenging**.
- By analyzing historical data, data-driven approaches can detect anomalies and predict failures with high accuracy.
- A **significant drawback** is the **need for extensive historical data** across various conditions and fault modes. Also **“black box” nature** of data-driven methods limits transparency.

Health Indicator: Hybrid

- Integration of the previous two approaches is increasingly seen to be a promising avenue for enhancing fault diagnosis in rotary machinery.
- Combining the deep physical insights of physics-based models with the adaptive and predictive capabilities of data-driven techniques.

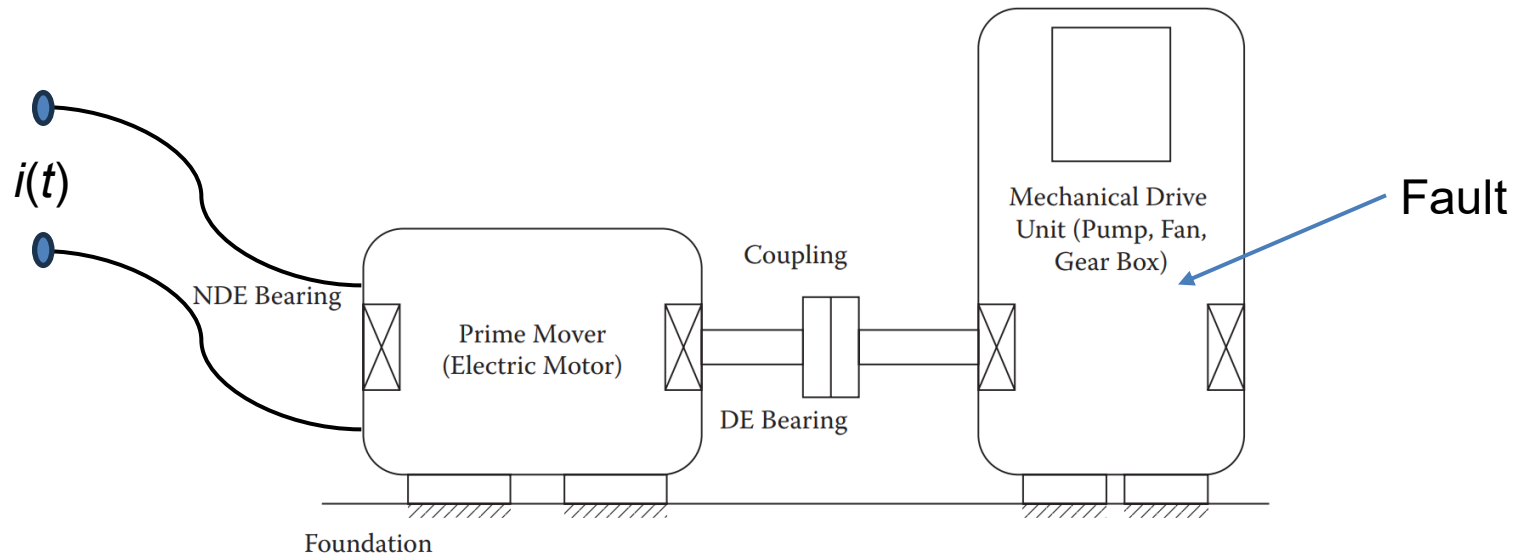


Bublil, T., Cohen, R., Kenett, R. S., & Bortman, J. (2025). Machine Health Indicators and Digital Twins. *Sensors*, 25(7), 2246.

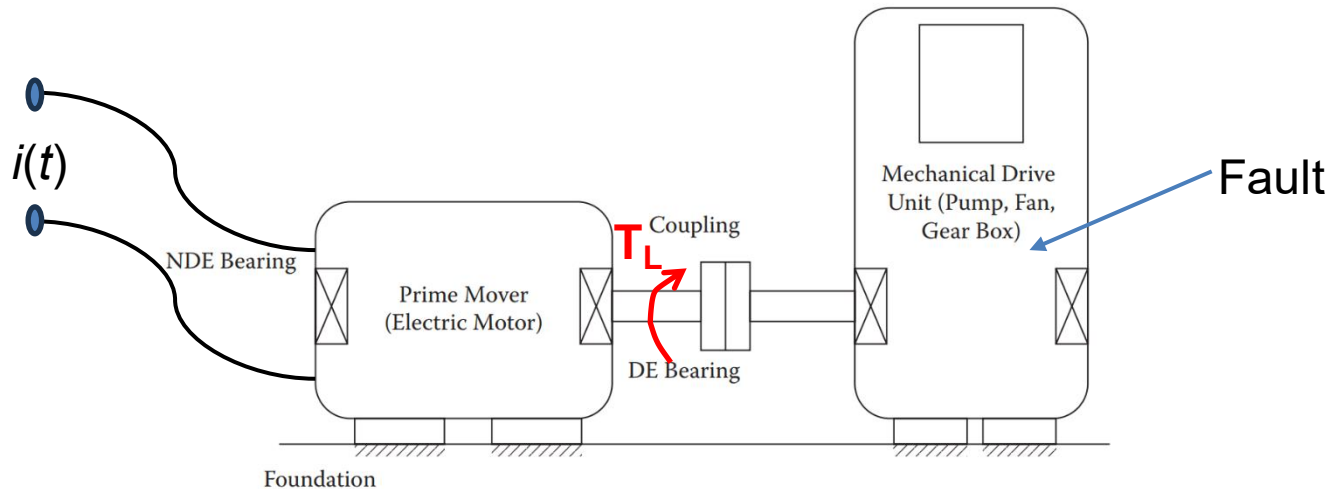
Motor Current Signature Analysis (MCSA)



- MCSA is a powerful technique for determining faults in mechanical systems that are **otherwise inaccessible** for health monitoring by conventional vibration analysis.
- It involves analyzing the current signal for predicting a failure event.
- **Monitored components need to be driven by an electric motor.**
- In this type of analysis, the motor itself acts as a transducer.
- Current is measured using Hall effect sensor, Rogowski coil etc.



MCSA: Theory



- Presence of **fault** in the mechanical element **will lead to a change in the Torque** (T_L)
- The load torque on the rotor will be a function of the input shaft speed and its modulated frequencies, **which will affect the electromagnetic field of the stator.**
- The stator will draw current according to the variation in the electromagnetic field.
- Presence of fault in the mechanical element will **modulate the current signal $i(t)$.**

Theory

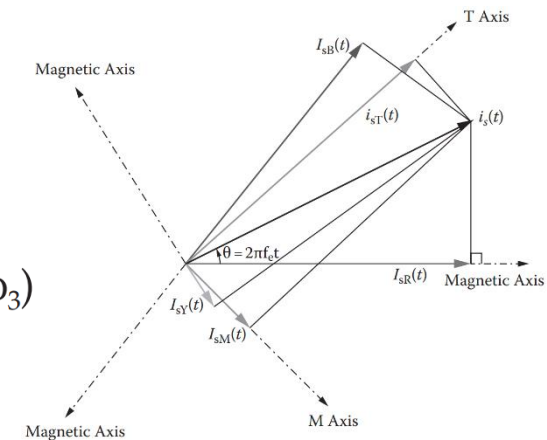


- The air-gap torque (T_L) in the induction motor consists of a constant (or average) torque and **some oscillatory torques due to torsional vibrations at some defect frequencies**

$$T_L = T_0 + \sum_i T_i$$

$$T = T_0 + T_1 \cos(2\pi f_1 t + \phi_1) + T_2 \cos(2\pi f_2 t + \phi_2) + T_3 \cos(2\pi f_3 t + \phi_3)$$

- Because of this the current will change.
- The current in any phase of the induction motor will have **two components**, **magnetizing** current component I_{sM} , which is in phase with flux vector; and a **torque** producing component I_{sT} , which is 90° ahead of the flux vector.



$$R - \text{phase current, } I_{sR} = I_{sM} \sin 2\pi f_s t + I_{sT} \cos 2\pi f_s t$$

Motor Current Signature Analysis (MCSA)



- Presence of defect will lead to

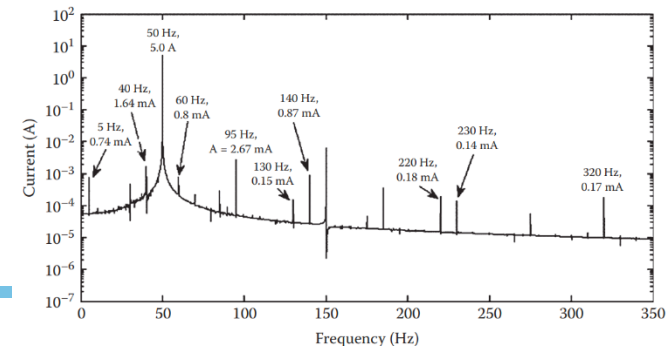
$$I_{sM} = I_{sM0} + A_{sM1} \sin(2\pi f_1 t + \phi_{M1}) + A_{sM2} \sin(2\pi f_2 t + \phi_{M2}) + A_{sM3} \sin(2\pi f_3 t + \phi_{M3})$$

$$I_{sT} = I_{sT0} + A_{sT1} \cos(2\pi f_1 t + \phi_{T1}) + A_{sT2} \cos(2\pi f_2 t + \phi_{T2}) + A_{sT3} \cos(2\pi f_3 t + \phi_{T3})$$

$$I_{sR} = I_{sM} \sin 2\pi f_s t + I_{sT} \sin 2\pi f_s t$$

- On simplification I_{sR} will become a function of f_s and f_1, f_2 etc.
- **Therefore, any mechanical defect, modulate the supply frequency.**
- For example, bearing defect can be identified by analysing the current spectrum as

$$f_{bng} = |f_s \pm m f_{i,o}|$$



Example

- In Motor Current Signature Analysis (MCSA), the load torque (T_L) of a motor connected to a faulty mechanical assembly is given by $T_L = 35 + 8 \cos(2\pi 10t) + 12 \cos(2\pi 5t)$ Nm. If the current is supplied at 230V/50 Hz, then:
- Identify the equation of current in one of the phases of the motor, say the R phase.
- Find the value of the current at time $t = 2$ seconds. Indicate proper unit.
- Identify the current equation under healthy state and draw its spectrum.

Given: $I_{sM} = A_{sM0} + \sum_{i=1}^N A_{sMi} \sin(2\pi f_i t + \phi_{sMi})$; $I_{sT} = A_{sT0} + \sum_{i=1}^N A_{sTi} \cos(2\pi f_i t + \phi_{sTi})$; $A_{sMi} = A_{sTi} = 1\text{A}$ for $i = 0$; $A_{sMi} = 1\text{A}$ for all $i = 1, \dots, N$; $\phi_{sMi} = 0$ for all $i = 1, \dots, N$; $A_{sTi} = 1\text{A}$ for all $i = 1, \dots, N$; and $\phi_{sTi} = 0$ for all $i = 1, \dots, N$.

Example

For a motor (230V/50 Hz) with a outer-race defective rolling-element bearing, the stator current components are modeled as

$$I_{sM}(t) = I_{sM0} + A_{sM1}\sin(2\pi f_1 t + \phi_{M1}) + A_{sM2}\sin(2\pi f_2 t + \phi_{M2})$$

$$I_{sT}(t) = I_{sT0} + A_{sT1}\cos(2\pi f_1 t + \phi_{T1}) + A_{sT2}\cos(2\pi f_2 t + \phi_{T2})$$

Here f_1 and f_2 are the 1st and 2nd harmonics of the fault characteristic frequency. The shaft rotational frequency is $f_r = 12 \text{ Hz}$.

The bearing parameters are:

Number of rolling elements: $n = 6$, Ball diameter: $d = 20 \text{ mm}$, Pitch diameter: $D = 45 \text{ mm}$, Contact angle: $\phi = 0^\circ$

Also assume:

$$\begin{aligned} I_{sM0} &= 5 \text{ A}, I_{sT0} = 2 \text{ A} \\ A_{sM1} &= 1.2 \text{ A}, A_{sT1} = 0.8 \text{ A} \\ A_{sM2} &= 0.6 \text{ A}, A_{sT2} = 0.4 \text{ A} \\ \phi_{M1} &= \phi_{M2} = \phi_{T1} = \phi_{T2} = 0 \end{aligned}$$

Find the current in R-phase and draw its spectrum with proper amplitudes

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- Reference: R3, Chapter 1, 2, 7, and 9